

MAT101R01 – ENGINEERING MATHEMATICS – I

(Common for all branches of Engineering) Semester I

Unit II: Matrix Algebra

Linear independence of vectors – Rank and row-echelon form – Eigen values and eigen vectors and their properties – Similarity and orthogonal transformations – Diagonalisation – Quadratic forms, nature and reduction to canonical form – Consistency of systems of linear equations – Applications.

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1 Vectors in matrix form

Consider a 2-D vector $\bar{X} = 2\vec{i} + 3\vec{j}$. This can be written in matrix form as

$$\bar{X} = \begin{pmatrix} 2 \\ 3 \end{pmatrix} \quad \text{or} \quad (2, 3),$$

i.e. either in column matrix or row matrix form. Suppose we have the 3-D vector $\bar{Y} = 2\vec{i} - \vec{j} + 4\vec{k}$; it can be represented in matrix form as $\bar{Y} = \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix}$ or $(2, -1, 4)$.

Hence any n -dimensional vector can be represented in matrix form as

$$X = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \quad \text{or} \quad (x_1, x_2, \dots, x_n).$$

1.1 Vector addition and scalar multiplication

Definition 1.1 (Vector addition and scalar multiplication)

Addition of two vectors $X = (x_1, x_2, \dots, x_n)$ and $Y = (y_1, y_2, \dots, y_n)$ is defined by

$$X + Y = (x_1, x_2, \dots, x_n) + (y_1, y_2, \dots, y_n) = (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n).$$

If $X = (x_1, \dots, x_n)$ is an n -dimensional vector and α is a scalar, then the scalar multiplication is defined by

$$\alpha X = \alpha(x_1, x_2, \dots, x_n) = (\alpha x_1, \alpha x_2, \dots, \alpha x_n).$$

1.2 Linear combination of vectors

Suppose $X_1 = (2, 3)$ and $X_2 = (1, -1)$, and $X_3 = X_1 - X_2 = (2, 3) - (1, -1) = (1, 4)$. Then we say X_3 is a linear combination of X_1 and X_2 , i.e.

$$X_3 = \alpha_1 X_1 + \alpha_2 X_2, \quad \text{where } \alpha_1 = 1, \alpha_2 = -1.$$

Definition 1.2 (Linear combination)

A vector X is said to be a *linear combination* of the vectors $X_1, X_2, \dots, X_n$ if it can be written as

$$X = \alpha_1 X_1 + \alpha_2 X_2 + \dots + \alpha_n X_n$$

for some scalars $\alpha_1, \dots, \alpha_n$.

Note

Any 2-D vector can be written as a linear combination of the 2-D unit vectors $(1, 0)$ and $(0, 1)$, i.e. $X = (\alpha_1, \alpha_2) = \alpha_1(1, 0) + \alpha_2(0, 1)$. Similarly any 3-D vector $(\alpha_1, \alpha_2, \alpha_3)$ can be written as a linear combination of the unit vectors $X_1 = (1, 0, 0)$, $X_2 = (0, 1, 0)$, $X_3 = (0, 0, 1)$:

$$(\alpha_1, \alpha_2, \alpha_3) = \alpha_1(1, 0, 0) + \alpha_2(0, 1, 0) + \alpha_3(0, 0, 1) = \alpha_1 X_1 + \alpha_2 X_2 + \alpha_3 X_3.$$

E.g. $(1, 2, 3) = 1(1, 0, 0) + 2(0, 1, 0) + 3(0, 0, 1)$.

1.3 Linearly independent and dependent vectors

Consider the first example $X_3 = X_1 - X_2 \Rightarrow X_1 - X_2 - X_3 = \bar{0}$, i.e. $\alpha_1 X_1 + \alpha_2 X_2 + \alpha_3 X_3 = \bar{0}$ with $\alpha_1 = 1$, $\alpha_2 = -1$, $\alpha_3 = -1$. Here X_1, X_2, X_3 are linearly *dependent* vectors.

Definition 1.3 (Linear dependence and independence)

A set of vectors $X_1, X_2, \dots, X_n$ is said to be **linearly dependent** if there exists a relation

$$\alpha_1 X_1 + \alpha_2 X_2 + \dots + \alpha_n X_n = \bar{0}$$

with some of the α_i 's non-zero.

Suppose the relation $\alpha_1 X_1 + \alpha_2 X_2 + \dots + \alpha_n X_n = \bar{0}$ is possible only with $\alpha_1 = \alpha_2 = \dots = \alpha_n = 0$ (all the α_i 's are zero). Then the vectors $X_1, X_2, \dots, X_n$ are said to be **linearly independent**.

Problem 1.1

Verify whether the vectors $(1, 2)$, $(1, -1)$, $(2, 1)$ are linearly dependent.

Solution. Let $X_1 = (1, 2)$, $X_2 = (1, -1)$, $X_3 = (2, 1)$ and consider $\alpha_1 X_1 + \alpha_2 X_2 + \alpha_3 X_3 = \bar{0}$:

$$\begin{aligned} \Rightarrow \alpha_1(1, 2) + \alpha_2(1, -1) + \alpha_3(2, 1) &= (0, 0) \\ \Rightarrow (\alpha_1, 2\alpha_1) + (\alpha_2, -\alpha_2) + (2\alpha_3, \alpha_3) &= (0, 0) \\ \Rightarrow (\alpha_1 + \alpha_2 + 2\alpha_3, 2\alpha_1 - \alpha_2 + \alpha_3) &= (0, 0), \end{aligned}$$

i.e.

$$\alpha_1 + \alpha_2 + 2\alpha_3 = 0 \quad (1), \quad 2\alpha_1 - \alpha_2 + \alpha_3 = 0 \quad (2).$$

By CMR (cross multiplication rule),

$$\frac{\alpha_1}{\begin{vmatrix} 1 & 2 \\ -1 & 1 \end{vmatrix}} = \frac{-\alpha_2}{\begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix}} = \frac{\alpha_3}{\begin{vmatrix} 1 & 1 \\ 2 & -1 \end{vmatrix}} = k \text{ (say)} \Rightarrow \frac{\alpha_1}{3} = \frac{-\alpha_2}{-3} = \frac{\alpha_3}{-3} = k \Rightarrow \frac{\alpha_1}{1} = \frac{\alpha_2}{1} = \frac{\alpha_3}{-1} = k,$$

so $\alpha_1 = k$, $\alpha_2 = k$, $\alpha_3 = -k$. Take $k = 1$: $\alpha_1 = 1$, $\alpha_2 = 1$, $\alpha_3 = -1$.

That is, $\alpha_1 X_1 + \alpha_2 X_2 + \alpha_3 X_3 = \bar{0}$ is possible with non-zero α_i 's. Hence the vectors X_1, X_2, X_3 are linearly dependent, and

$$X_1 + X_2 - X_3 = \bar{0} \Rightarrow \boxed{X_3 = X_1 + X_2}$$

■

1.4 Checking linear independence and dependence of vectors

Follow any one of the following methods.

(1) **Direct definition.** Form the relation $\alpha_1 X_1 + \alpha_2 X_2 + \dots + \alpha_n X_n = \bar{0}$ and identify the values of $\alpha_1, \alpha_2, \dots, \alpha_n$.

(2) **Determinant value of the matrix.** Applicable for 3 vectors with 3 components (or 2 vectors with 2 components).

(i) Construct a 3×3 (or 2×2) matrix A where each row is a vector.

(ii) Identify the value of the determinant of A .

(iii) If $|A| = 0$, the vectors are linearly dependent; if $|A| \neq 0$, the vectors are linearly independent.

(3) **Row-echelon form.**

(i) A matrix A is constructed through the given vectors, where each row of the matrix is a vector.

(ii) Apply row (or column) transformations to transform A into its *row-echelon form* (converting the elements below the principal diagonal to zero).

(iii) Now consider the number of non-zero rows in the row-echelon form; let it be r .

(iv) If $r < n$ (the number of vectors), the vectors are linearly dependent. If $r = n$, the vectors are linearly independent.

Problem 1.2

Verify whether the vectors $(1, 2, -1)$, $(2, 3, 5)$ and $(1, 1, 6)$ are linearly independent or dependent. If dependent, identify the linear relationship among them.

Solution. Given vectors $X_1 = (1, 2, -1)$, $X_2 = (2, 3, 5)$, $X_3 = (1, 1, 6)$. The matrix formed from them is

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 3 & 5 \\ 1 & 1 & 6 \end{pmatrix}, \quad |A| = 1(18 - 5) - 2(12 - 5) - 1(2 - 3) = 13 - 14 + 1 = 0.$$

$\Rightarrow$ The given vectors are linearly dependent.

To identify the relationship among them, we convert A into its row-echelon form. Apply $R_2 \rightarrow R_2 - 2R_1$, $R_3 \rightarrow R_3 - R_1$:

$$A \sim \begin{pmatrix} 1 & 2 & -1 \\ 0 & -1 & 7 \\ 0 & -1 & 7 \end{pmatrix}.$$

Apply $R_3 \rightarrow R_3 - R_2$:

$$A \sim \begin{pmatrix} 1 & 2 & -1 \\ 0 & -1 & 7 \\ 0 & 0 & 0 \end{pmatrix}, \quad \text{which is the row-echelon form.}$$

Here the number of non-zero rows is $r = 2 < 3 = n$ (number of given vectors) $\Rightarrow$ the vectors are linearly dependent.

To get the linear relationship: in the row-echelon form $R_3 = 0$, i.e.

$$\begin{aligned} R_3 - R_2 &= 0 \quad (\text{previous transformation}) \\ \Rightarrow (R_3 - R_1) - (R_2 - 2R_1) &= 0 \quad (\text{previous transformations}) \\ \Rightarrow R_3 - R_2 + R_1 &= 0, \quad \text{which represents the original matrix } A. \end{aligned}$$

But in the original matrix each row is a vector, so we have

$$\boxed{X_3 - X_2 + X_1 = 0} \quad \text{or} \quad \boxed{X_2 = X_1 + X_3},$$

which is the linear relationship among them. ■

Problem 1.3

Show that the vectors $X_1 = (1, -1, -2, -4)$, $X_2 = (2, 3, -1, -1)$, $X_3 = (3, 1, 3, -2)$ and $X_4 = (6, 3, 0, -7)$ are linearly dependent and find their relationship.

Solution. Let us consider the matrix A formed from the vectors:

$$A = \begin{pmatrix} 1 & -1 & -2 & -4 \\ 2 & 3 & -1 & -1 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{pmatrix}.$$

Apply $R_2 \rightarrow R_2 - 2R_1$, $R_3 \rightarrow R_3 - 3R_1$, $R_4 \rightarrow R_4 - 6R_1$:

$$A \sim \begin{pmatrix} 1 & -1 & -2 & -4 \\ 0 & 5 & 3 & 7 \\ 0 & 4 & 9 & 10 \\ 0 & 9 & 12 & 17 \end{pmatrix}.$$

Apply $R_2 \rightarrow \frac{1}{5}R_2$:

$$A \sim \begin{pmatrix} 1 & -1 & -2 & -4 \\ 0 & 1 & 3/5 & 7/5 \\ 0 & 4 & 9 & 10 \\ 0 & 9 & 12 & 17 \end{pmatrix}.$$

Apply $R_3 \rightarrow R_3 - 4R_2$, $R_4 \rightarrow R_4 - 9R_2$ (e.g. $9 - \frac{12}{5} = \frac{33}{5}$, $10 - \frac{28}{5} = \frac{22}{5}$, $12 - \frac{27}{5} = \frac{33}{5}$, $17 - \frac{63}{5} = \frac{22}{5}$):

$$A \sim \begin{pmatrix} 1 & -1 & -2 & -4 \\ 0 & 1 & 3/5 & 7/5 \\ 0 & 0 & 33/5 & 22/5 \\ 0 & 0 & 33/5 & 22/5 \end{pmatrix}.$$

Apply $R_4 \rightarrow R_4 - R_3$:

$$A \sim \begin{pmatrix} 1 & -1 & -2 & -4 \\ 0 & 1 & 3/5 & 7/5 \\ 0 & 0 & 33/5 & 22/5 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad \text{the row-echelon form of } A.$$

Here the number of non-zero rows = 3 < 4 = the number of given vectors. $\therefore$ The given vectors are linearly dependent.

To get the linear relationship, we have $R_4 = 0$ in the row-echelon form:

$$\begin{aligned} \Rightarrow R_4 - R_3 &= 0 \quad (\text{previously}) \\ \Rightarrow (R_4 - 9R_2) - (R_3 - 4R_2) &= 0 \quad (\text{previously}) \\ \Rightarrow R_4 - R_3 - 5R_2 &= 0 \\ \Rightarrow R_4 - R_3 - 5\left(\frac{1}{5}R_2\right) &= 0 \quad (\text{previously}) \Rightarrow R_4 - R_3 - R_2 = 0 \\ \Rightarrow (R_4 - 6R_1) - (R_3 - 3R_1) - (R_2 - 2R_1) &= 0 \\ \Rightarrow R_4 - R_3 - R_2 - R_1 &= 0, \quad \text{the original matrix relation.} \end{aligned}$$

But in A each row is a vector:

$$\Rightarrow \boxed{X_4 - X_3 - X_2 - X_1 = 0} \quad \text{or} \quad \boxed{X_4 = X_1 + X_2 + X_3}$$

is the linear relationship among them. ■

Exercises.

- (4) Show that the vectors $X_1 = (1, 1, 2)$, $X_2 = (1, 2, 5)$ and $X_3 = (5, 3, 4)$ are linearly dependent. Also find the linear relationship among them. **Ans:** $7X_1 - 2X_2 - X_3 = 0$.
- (5) Check whether the vectors $X_1 = (2, -2, 1)$, $X_2 = (1, 4, -1)$ and $X_3 = (4, 6, -3)$ are linearly independent. **Ans:** $|A| = -20 \neq 0 \Rightarrow$ independent.
- (6) Show that the following sets of vectors are linearly dependent. Find their relationship.
 - (i) $X_1 = (1, 2, -1, 3)$, $X_2 = (0, -2, 1, -1)$, $X_3 = (2, 2, -1, 5)$. **Ans:** $2X_1 + X_2 = X_3$.
 - (ii) $X_1 = (1, -2, 4, 1)$, $X_2 = (1, 0, 6, -5)$, $X_3 = (2, -3, 9, -1)$ and $X_4 = (2, -5, 7, 5)$. **Ans:** $3X_1 + X_2 - 2X_3 = 0$, $5X_1 - X_2 - 2X_4 = 0$.
- (7) Show that the vectors $X_1 = (1, -2, -3, -2, 1)$, $X_2 = (3, -2, 0, -1, -7)$, $X_3 = (0, 1, 2, 1, -6)$ and $X_4 = (0, 2, 2, 1, -5)$ are linearly independent.

2 Eigen values and eigen vectors

Consider the vector $X = \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$ and the linear transformation (L.T.)

$$\bar{X} = 4x - 3y, \quad \bar{Y} = 2x + 3y \Rightarrow \begin{pmatrix} \bar{X} \\ \bar{Y} \end{pmatrix} = \begin{pmatrix} 4 & -3 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = A \begin{pmatrix} x \\ y \end{pmatrix},$$

which takes the old vector to a new vector. If the new vector satisfies $\begin{pmatrix} \bar{X} \\ \bar{Y} \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$, then $\begin{pmatrix} x \\ y \end{pmatrix}$ is called an eigen vector and λ an eigen value.

In general, we have the vector X_1 and apply the matrix of L.T. A on X_1 ; we get a new vector X_2 such that $X_2 = AX_1$. Then if $AX_1 = \lambda X_1$, i.e. $X_2 = \lambda X_1$ where λ is a scalar, the vector X_1 is called an eigen vector of the matrix of L.T. A and λ is called an eigen value for the matrix of L.T. A . That is, the vectors which do not change their direction after applying a force or L.T. are called eigen vectors, and the scalar λ is called the eigen value of the matrix of L.T.

Definition 2.1 (Eigen value and eigen vector)

A non-zero vector X is said to be an **eigen vector** for the matrix of linear transformation A if it satisfies $AX = \lambda X$, where λ is a scalar. The scalar λ is called an **eigen value** of A .

2.1 Identifying eigen values and eigen vectors for a matrix of L.T. A

Suppose X is an eigen vector of a matrix of L.T. A . Then it satisfies

$$\begin{aligned} AX = \lambda X &\Rightarrow AX = \lambda IX \Rightarrow AX - \lambda IX = 0 \Rightarrow (A - \lambda I)X = 0 \\ &\Rightarrow \begin{pmatrix} a_{11} - \lambda & a_{12} & a_{13} \\ a_{21} & a_{22} - \lambda & a_{23} \\ a_{31} & a_{32} & a_{33} - \lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad (\text{for example}) \end{aligned}$$

will be a system of homogeneous type equations. The above system of equations possesses a non-zero solution only when $|A - \lambda I| = 0$.

Definition 2.2 (Characteristic equation)

The equation $|A - \lambda I| = 0$ is called the **characteristic equation** of A , and its roots are called the **eigen values** of A . When we use the relation $AX = \lambda X$ for each eigen value λ , we get the respective eigen vectors.

I.e.

- (i) For the given matrix A , construct the characteristic equation $|A - \lambda I| = 0$.
- (ii) Identify the roots of the characteristic equation $|A - \lambda I| = 0$; the roots are called the eigen values of A .
- (iii) For each eigen value λ of A , use the relation $AX = \lambda X$; we get the respective eigen vectors.

Result 2.1 (Property)

- (i) Sum of the eigen values of A = Sum of the diagonal elements of A .
- (ii) Product of the eigen values of A = $|A|$ = determinant value of A .

Proof. Let us prove this property by taking a 2×2 matrix $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$. Then the characteristic equation of A is $|A - \lambda I| = 0$:

$$\begin{aligned} &\Rightarrow \begin{vmatrix} a_{11} - \lambda & a_{12} \\ a_{21} & a_{22} - \lambda \end{vmatrix} = 0 \Rightarrow (a_{11} - \lambda)(a_{22} - \lambda) - a_{12}a_{21} = 0 \\ &\Rightarrow (\lambda - a_{11})(\lambda - a_{22}) - a_{12}a_{21} = 0 \Rightarrow \lambda^2 - (a_{11} + a_{22})\lambda + (a_{11}a_{22} - a_{12}a_{21}) = 0, \end{aligned} \quad (1)$$

which is the characteristic equation of A . Since A is a matrix of order 2, let λ_1, λ_2 be the eigen values of A . Already we know that the eigen values are the roots of the characteristic equation, so the characteristic equation of A is given by (recall: roots $\alpha, \beta \Rightarrow (x - \alpha)(x - \beta) = 0$)

$$(\lambda - \lambda_1)(\lambda - \lambda_2) = 0 \Rightarrow \lambda^2 - (\lambda_1 + \lambda_2)\lambda + \lambda_1\lambda_2 = 0. \quad (2)$$

By (1) and (2), we see that $a_{11} + a_{22} = \lambda_1 + \lambda_2$, i.e.

$$\boxed{\text{Sum of the diagonal elements of } A = \text{Sum of the eigen values of } A}$$

and $a_{11}a_{22} - a_{12}a_{21} = \lambda_1\lambda_2$, i.e. det. value of A = product of eigen values of A :

$$\boxed{|A| = \text{Product of the eigen values of } A}$$

Note

(i) For any 2×2 matrix A , its characteristic equation is given by

$$\lambda^2 - (\text{Sum of the diagonal elements of } A)\lambda + |A| = 0.$$

(ii) For any 3×3 matrix A , its characteristic equation is given by

$$\lambda^3 - (\text{Sum of the diagonal elements of } A)\lambda^2 + (\text{Sum of the det. minors through the diagonals of } A)\lambda - |A| = 0,$$

where the sum of the minors of the diagonal elements is $\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} + \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}.$

2.2 Problems**Problem 2.1**

Find the eigen values and eigen vectors for the matrix of L.T. $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}.$

Solution. The characteristic equation is $|A - \lambda I| = 0$ and it is given by $\lambda^2 - (\text{sum of diagonal elements of } A)\lambda + |A| = 0$

$$\Rightarrow \lambda^2 - 2\lambda + (1 - 4) = 0 \Rightarrow \boxed{\lambda^2 - 2\lambda - 3 = 0} \text{ is the char. eqn. of } A$$

$$(\text{Directly: } \begin{vmatrix} 1 - \lambda & 2 \\ 2 & 1 - \lambda \end{vmatrix} = 0 \Rightarrow (1 - \lambda)^2 - 4 = 0 \Rightarrow 1 - 2\lambda + \lambda^2 - 4 = 0.)$$

$$\Rightarrow (\lambda - 3)(\lambda + 1) = 0 \Rightarrow \boxed{\lambda = -1, 3} \text{ are the eigen values of } A.$$

Eigen vectors.

(i) For $\lambda = -1$: $AX = \lambda X$

$$\Rightarrow \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = -1 \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{cases} x + 2y = -x \\ 2x + y = -y \end{cases} \Rightarrow \begin{cases} 2x + 2y = 0 \\ 2x + 2y = 0 \end{cases}$$

(both are the same equation) $\Rightarrow x + y = 0 \Rightarrow x = -y \Rightarrow \frac{x}{-1} = \frac{y}{1} = k$ (say), i.e. $x = -k$, $y = k$. Take $k = 1$:
 $x = -1$, $y = 1 \Rightarrow$ the eigen vector for $\lambda = -1$ is $\begin{pmatrix} -1 \\ 1 \end{pmatrix}.$

(ii) For $\lambda = 3$: $AX = \lambda X$

$$\Rightarrow \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 3 \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{cases} x + 2y = 3x \\ 2x + y = 3y \end{cases} \Rightarrow \begin{cases} -2x + 2y = 0 \\ 2x - 2y = 0 \end{cases}$$

The above two equations are reduced to $x - y = 0 \Rightarrow x = y \Rightarrow \frac{x}{1} = \frac{y}{1} = k$ (say). Take $k = 1$: $x = 1$, $y = 1 \Rightarrow$ the eigen vector for $\lambda = 3$ is $\begin{pmatrix} 1 \\ 1 \end{pmatrix}.$

I.e. the eigen values are $\lambda = -1, 3$ and the eigen vectors are $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 1 \end{pmatrix}.$ ■

Problem 2.2

Find the eigen values and eigen vectors of the matrix of L.T. $A = \begin{pmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{pmatrix}.$

Solution. The characteristic equation is given by $|A - \lambda I| = 0$, i.e.

$$\lambda^3 - (\text{sum of the diagonal elts. of } A)\lambda^2 + (\text{sum of the det. minors thro' the diagonals of } A)\lambda - |A| = 0$$

$$\begin{aligned}
&\Rightarrow \lambda^3 - (7)\lambda^2 + \left[\begin{vmatrix} 5 & 1 \\ 1 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 3 \\ 3 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 1 \\ 1 & 5 \end{vmatrix} \right] \lambda - [1(4) - 1(-2) + 3(-14)] = 0 \\
&\Rightarrow \lambda^3 - 7\lambda^2 + (4 - 8 + 4)\lambda - (4 + 2 - 42) = 0 \\
&\Rightarrow \lambda^3 - 7\lambda^2 + 36 = 0 \quad \text{is the char. eqn. of } A.
\end{aligned}$$

If $\lambda = 1$: $1 - 7 + 36 \neq 0$; $\lambda = -1$: $-1 - 7 + 36 \neq 0$; $\lambda = 2$: $8 - 28 + 36 \neq 0$; $\lambda = -2$: $-8 - 28 + 36 = 0$.

By substitution, $\lambda = -2$ is one root and the remaining roots are obtained by synthetic division:

$$\begin{array}{r|rrrr}
-2 & 1 & -7 & 0 & 36 \\
& & 0 & -2 & 18 & -36 \\
\hline
& 1 & -9 & 18 & 0
\end{array}$$

$\therefore$ the remaining factor is $\lambda^2 - 9\lambda + 18 = 0 \Rightarrow (\lambda - 6)(\lambda - 3) = 0 \Rightarrow \lambda = 3, 6$.

$\therefore$ Eigen values are $\boxed{\lambda = -2, 3, 6}$

Eigen vectors.

(i) For $\lambda = -2$: we have $AX = \lambda X$

$$\Rightarrow \begin{pmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = -2 \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \Rightarrow \begin{cases} x_1 + x_2 + 3x_3 = -2x_1 \\ x_1 + 5x_2 + x_3 = -2x_2 \\ 3x_1 + x_2 + x_3 = -2x_3 \end{cases} \Rightarrow \begin{cases} 3x_1 + x_2 + 3x_3 = 0 \\ x_1 + 7x_2 + x_3 = 0 \\ 3x_1 + x_2 + 3x_3 = 0 \end{cases}$$

I.e. the above three equations are reduced to

$$3x_1 + x_2 + 3x_3 = 0 \quad (1), \quad x_1 + 7x_2 + x_3 = 0 \quad (2).$$

Using CMR for (1) and (2):

$$\frac{x_1}{\begin{vmatrix} 1 & 3 \\ 7 & 1 \end{vmatrix}} = \frac{-x_2}{\begin{vmatrix} 3 & 3 \\ 1 & 1 \end{vmatrix}} = \frac{x_3}{\begin{vmatrix} 3 & 1 \\ 1 & 7 \end{vmatrix}} = k \Rightarrow \frac{x_1}{1-21} = \frac{-x_2}{3-3} = \frac{x_3}{21-1} \Rightarrow \frac{x_1}{-20} = \frac{x_2}{0} = \frac{x_3}{20} \Rightarrow \frac{x_1}{1} = \frac{x_2}{0} = \frac{x_3}{-1} = k.$$

Take $k = 1$: $x_1 = 1$, $x_2 = 0$, $x_3 = -1$. Eigen vector for $\lambda = -2$ is $X_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$.

(ii) For $\lambda = 3$: $AX = 3X$

$$\Rightarrow -2x_1 + x_2 + 3x_3 = 0 \quad (1), \quad x_1 + 2x_2 + x_3 = 0 \quad (2), \quad 3x_1 + x_2 - 2x_3 = 0 \quad (3).$$

Apply CMR for (1) and (2):

$$\frac{x_1}{1-6} = \frac{-x_2}{-2-3} = \frac{x_3}{-4-1} = k \Rightarrow \frac{x_1}{-5} = \frac{x_2}{5} = \frac{x_3}{-5} \Rightarrow \frac{x_1}{1} = \frac{x_2}{-1} = \frac{x_3}{1} = k.$$

Take $k = 1$: $x_1 = 1$, $x_2 = -1$, $x_3 = 1 \Rightarrow$ eigen vector for $\lambda = 3$ is $X_2 = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$.

(iii) For $\lambda = 6$: $AX = 6X$

$$\Rightarrow -5x_1 + x_2 + 3x_3 = 0 \quad (1), \quad x_1 - x_2 + x_3 = 0 \quad (2), \quad 3x_1 + x_2 - 5x_3 = 0 \quad (3).$$

Apply CMR for (1) and (2):

$$\frac{x_1}{1+3} = \frac{-x_2}{-5-3} = \frac{x_3}{5-1} = k \Rightarrow \frac{x_1}{4} = \frac{x_2}{8} = \frac{x_3}{4} \Rightarrow \frac{x_1}{1} = \frac{x_2}{2} = \frac{x_3}{1} = k.$$

Take $k = 1$: $x_1 = 1$, $x_2 = 2$, $x_3 = 1 \Rightarrow$ eigen vector for $\lambda = 6$ is $X_3 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$.

I.e. we have the eigen values $-2, 3, 6$ and eigen vectors $\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$. ■

Exercise. Find the eigen values and eigen vectors of the matrix of L.T. $A = \begin{pmatrix} 1 & 2 & -3 \\ 0 & 3 & 2 \\ 0 & 0 & -2 \end{pmatrix}$.

(Char. eqn. $\lambda^3 - 2\lambda^2 - 5\lambda + 6 = 0$; $\lambda = 1, -2, 3$.)

Note

(i) For a matrix of L.T., if all the eigen values are *distinct*, then the respective eigen vectors are *linearly independent*.

(ii) Suppose some of the eigen values are equal; then the respective eigen vectors may or may not be linearly independent.

The following problems illustrate these ideas.

Problem 2.3

Find the eigen values and eigen vectors of the matrix $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$.

Solution. The characteristic equation is $|A - \lambda I| = 0$, or

$$\lambda^3 - (\text{sum of the diagonal elts. of } A)\lambda^2 + (\text{sum of the det. minors thro' the diagonals of } A)\lambda - |A| = 0$$

. Here the sum of diagonal elements is 0, the sum of minors is $(0 - 1) + (0 - 1) + (0 - 1) = -3$ and $|A| = 0 - 1(0 - 1) + 1(1 - 0) = 2$:

$$\Rightarrow \boxed{\lambda^3 - 3\lambda - 2 = 0} \Rightarrow (\lambda + 1)^2(\lambda - 2) = 0.$$

Then the eigen values are $\boxed{\lambda = -1, -1, 2}$.

Eigen vectors.

(i) For $\lambda = 2$: $AX = \lambda X$

$$\Rightarrow \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 2 \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$\Rightarrow -2x_1 + x_2 + x_3 = 0 \quad (1), \quad x_1 - 2x_2 + x_3 = 0 \quad (2), \quad x_1 + x_2 - 2x_3 = 0 \quad (3).$$

Apply CMR on (1) and (2): $\frac{x_1}{1+2} = \frac{-x_2}{-2-1} = \frac{x_3}{4-1} \Rightarrow \frac{x_1}{1} = \frac{x_2}{1} = \frac{x_3}{1} = k$, and take $k = 1 \Rightarrow$ eigen vector

$$X_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

(ii) For $\lambda = -1$: $AX = \lambda X \Rightarrow$

$$\begin{cases} x_1 + x_2 + x_3 = 0 \\ x_1 + x_2 + x_3 = 0 \\ x_1 + x_2 + x_3 = 0 \end{cases}$$

All the above three equations are the same, representing a single equation, i.e. $x_1 + x_2 + x_3 = 0$ (1).

Assume $x_3 = 0$: (1) $\Rightarrow x_1 + x_2 = 0 \Rightarrow x_1 = -x_2 \Rightarrow \frac{x_1}{-1} = \frac{x_2}{1} = k$ (say). Take $k = 1$: $x_1 = -1$, $x_2 = 1 \Rightarrow$ eigen

$$\text{vector is } X_2 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}.$$

(iii) Again for $\lambda = -1$: by $AX = \lambda X$ we get $x_1 + x_2 + x_3 = 0$ (2). Now we assume $x_2 = 0$: $x_1 + x_3 = 0 \Rightarrow x_1 =$

$$-x_3 \Rightarrow \frac{x_1}{-1} = \frac{x_3}{1} = k \text{ (say)}. \text{ Take } k = 1: x_1 = -1, x_3 = 1 \Rightarrow \text{eigen vector } X_3 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}.$$

$\therefore$ We have the three independent eigen vectors $X_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$, $X_2 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$, $X_3 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$. ■

Note

In the above problem, suppose we assume $x_1 = 0$ for $\lambda = -1$. We get $x_2 + x_3 = 0 \Rightarrow x_2 = -x_3 \Rightarrow \frac{x_2}{-1} = \frac{x_3}{1} = k$, giving $X_4 = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}$. But this $X_4 = X_3 - X_2$, so there is no necessity of finding this vector.

Problem 2.4

Find the eigen values and eigen vectors of the matrix $A = \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$.

Solution. The characteristic equation is given by $|A - \lambda I| = 0$, or

$$\lambda^3 - (\text{sum of the diagonal elts.})\lambda^2 + (\text{sum of the det. minors thro' the diagonals})\lambda - |A| = 0$$

$$\Rightarrow \lambda^3 - 12\lambda^2 + (8 + 14 + 14)\lambda - 32 = 0 \Rightarrow \lambda^3 - 12\lambda^2 + 36\lambda - 32 = 0,$$

which is the char. eqn. By substitution $\lambda = 2$ is one root ($8 - 48 + 72 - 32 = 0$), and the remaining roots are $\lambda = 2, 8$. Then the eigen values are $\boxed{\lambda = 2, 2, 8}$.

Eigen vectors.

(i) For $\lambda = 8$: $AX = \lambda X$

$$\Rightarrow \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 8 \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \Rightarrow \begin{cases} -2x_1 - 2x_2 + 2x_3 = 0 & (1) \\ -2x_1 - 5x_2 - x_3 = 0 & (2) \\ 2x_1 - x_2 - 5x_3 = 0 & (3) \end{cases}$$

All the above equations are distinct. By CMR on (1) and (2):

$$\frac{x_1}{2+10} = \frac{-x_2}{2+4} = \frac{x_3}{10-4} = k \Rightarrow \frac{x_1}{12} = \frac{x_2}{-6} = \frac{x_3}{6} \Rightarrow \frac{x_1}{2} = \frac{x_2}{-1} = \frac{x_3}{1} = k.$$

Take $k = 1$: $x_1 = 2$, $x_2 = -1$, $x_3 = 1$; the eigen vector is $X_1 = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$.

(ii) For $\lambda = 2$: $AX = \lambda X$

$$\Rightarrow \begin{cases} 4x_1 - 2x_2 + 2x_3 = 0 \\ -2x_1 + x_2 - x_3 = 0 \\ 2x_1 - x_2 + x_3 = 0 \end{cases}$$

The above three equations are the same and it will be $\boxed{2x_1 - x_2 + x_3 = 0}$ (1). If we assume $x_3 = 0$: (1)

$\Rightarrow 2x_1 - x_2 = 0 \Rightarrow 2x_1 = x_2 \Rightarrow \frac{x_1}{1} = \frac{x_2}{2} = k$ (say). Taking $k = 1$: $x_1 = 1$, $x_2 = 2 \Rightarrow$ eigen vector $X_2 = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$.

(iii) Again for $\lambda = 2$: $AX = \lambda X$ provides $2x_1 - x_2 + x_3 = 0$. Here, if we assume $x_2 = 0$, then $2x_1 + x_3 = 0 \Rightarrow 2x_1 = -x_3 \Rightarrow \frac{x_1}{-1} = \frac{x_3}{2} = k$ (say). Take $k = 1$: $x_1 = -1$, $x_3 = 2 \Rightarrow$ eigen vector $X_3 = \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix}$.

I.e. the eigen vectors $X_2 = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$ and $X_3 = \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix}$ are the two distinct (independent) eigen vectors for $\lambda = 2$.

[Suppose we assume $x_1 = 0$ in (1) of $AX = \lambda X$: $-x_2 + x_3 = 0 \Rightarrow x_2 = x_3 \Rightarrow \frac{x_2}{1} = \frac{x_3}{1} = k$, giving the eigen vector

$X_4 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \frac{1}{2}(X_2 + X_3)$. This X_4 is not linearly independent, so there is no necessity of finding X_4 .]

∴ For this problem, even though two eigen values are the same, we are able to identify three independent eigen vectors; they are $X_1 = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$, $X_2 = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$ and $X_3 = \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix}$. ■

Problem 2.5

Find the eigen values and eigen vectors for the matrix $A = \begin{pmatrix} 2 & -2 & 2 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{pmatrix}$.

Solution. Char. eqn.: $\lambda^3 - 2\lambda^2 - 4\lambda + 8 = 0 \Rightarrow (\lambda - 2)^2(\lambda + 2) = 0$. Eigen values are $\boxed{\lambda = 2, 2, -2}$.

Eigen vectors.

(i) For $\lambda = -2$: $AX = \lambda X \Rightarrow \begin{pmatrix} 2 & -2 & 2 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = -2 \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$, i.e. $4x_1 - 2x_2 + 2x_3 = 0$, $x_1 + 3x_2 + x_3 = 0$, $x_1 + 3x_2 + x_3 = 0$. By CMR on the first two, $\frac{x_1}{-2-6} = \frac{-x_2}{4-2} = \frac{x_3}{12+2} \Rightarrow \frac{x_1}{-8} = \frac{x_2}{-2} = \frac{x_3}{14} \Rightarrow \frac{x_1}{4} = \frac{x_2}{1} = \frac{x_3}{-7} \Rightarrow$ eigen vector $X_1 = \begin{pmatrix} 4 \\ 1 \\ -7 \end{pmatrix}$.

(ii) For $\lambda = 2$: $AX = \lambda X$

$$\Rightarrow 0x_1 - 2x_2 + 2x_3 = 0 \text{ (1)}, \quad x_1 - x_2 + x_3 = 0 \text{ (2)}, \quad x_1 + 3x_2 - 3x_3 = 0 \text{ (3)}.$$

All the above three equations are distinct. ∴ Apply CMR on (1) and (2):

$$\frac{x_1}{-2+2} = \frac{-x_2}{0-2} = \frac{x_3}{0+2} = k \Rightarrow \frac{x_1}{0} = \frac{x_2}{2} = \frac{x_3}{2} \Rightarrow \frac{x_1}{0} = \frac{x_2}{1} = \frac{x_3}{1} = k.$$

Take $k = 1$: $x_1 = 0$, $x_2 = 1$, $x_3 = 1 \Rightarrow$ eigen vector $X_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$.

(iii) Again for $\lambda = 2$: by $AX = \lambda X$ we get three distinct equations, ∴ $X_3 = X_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$.

∴ Here the eigen vectors are *not* linearly independent. ■

Note

From the above two problems, we see that

- (i) For equal eigen values, if the respective equations from $AX = \lambda X$ reduce to a single equation, then we are able to identify independent eigen vectors for the same eigen values.
- (ii) If the equations from $AX = \lambda X$ are not reduced to a single equation (i.e. at least two equations are distinct), then we cannot find independent eigen vectors.

Exercise. Check whether the eigen vectors of the matrix $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ are linearly independent.

3 Diagonalization of a matrix

In this section we are finding the diagonal matrix D for the given square matrix A . Here both A and D are *similar* matrices. For similar matrices we have (i) the same eigen values, (ii) the same determinant value, (iii) $\text{Rank}(A) = \text{Rank}(D)$ — totally they have the same characteristics. Hence the diagonal matrix D will give all the necessary information about A .

The following transformations are used to get the diagonalization of a matrix:

- (1) **Similarity transformation** [used for any square matrix A];
- (2) **Orthogonal transformation** [applicable for only real symmetric matrices].

3.1 Similarity transformation

Result 3.1 (Steps in the similarity transformation)

- (i) For the given matrix A , identify its eigen values and eigen vectors; let them be $\lambda_1, \lambda_2, \lambda_3$ and X_1, X_2, X_3 .
- (ii) Construct the *modal matrix* B such that $B = (X_1 \ X_2 \ X_3)$, where X_1, X_2, X_3 are the eigen vectors of A .
- (iii) Identify B^{-1} using $B^{-1} = \frac{\text{adj}(B)}{|B|}$.
- (iv) Form the product of matrices $B^{-1}AB$; this product will be a diagonal matrix D and the elements of D are the eigen values of A .
- (v) Now A and D are similar matrices.

Problem 3.1

Diagonalize the matrix $A = \begin{pmatrix} 1 & 2 & -3 \\ 0 & 3 & 2 \\ 0 & 0 & -2 \end{pmatrix}$ using the similarity transformation.

Solution. The characteristic equation of A is $|A - \lambda I| = 0$, or

$$\lambda^3 - (\text{sum of the diagonal elts.})\lambda^2 + (\text{sum of the det. minors thro' the diagonals})\lambda - |A| = 0$$

$$\Rightarrow \lambda^3 - (2)\lambda^2 + (-6 + (-2) + 3)\lambda - (1(-6) + 0 + 0) = 0 \Rightarrow \lambda^3 - 2\lambda^2 - 5\lambda + 6 = 0.$$

$\lambda = 1$ is one root; by synthetic division

$$\begin{array}{r|rrrr} 1 & 1 & -2 & -5 & 6 \\ & 0 & 1 & -1 & -6 \\ \hline & 1 & -1 & -6 & 0 \end{array} \Rightarrow \lambda^2 - \lambda - 6 = 0 \Rightarrow (\lambda - 3)(\lambda + 2) = 0 \Rightarrow \lambda = -2, 3.$$

$\therefore$ Eigen values are $\boxed{\lambda = 1, -2, 3}$.

Eigen vectors. (i) For $\lambda = 1$: $AX = \lambda X$ gives $2x_2 - 3x_3 = 0$, $2x_2 + 2x_3 = 0$, $-3x_3 = 0 \Rightarrow x_2 = x_3 = 0$, x_1 arbitrary $\Rightarrow X_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$.

(ii) For $\lambda = -2$: $3x_1 + 2x_2 - 3x_3 = 0$, $5x_2 + 2x_3 = 0$. By CMR, $\frac{x_1}{4+15} = \frac{-x_2}{6-0} = \frac{x_3}{15-0} \Rightarrow X_2 = \begin{pmatrix} 19 \\ -6 \\ 15 \end{pmatrix}$.

(iii) For $\lambda = 3$: $-2x_1 + 2x_2 - 3x_3 = 0$, $2x_3 = 0$, $-5x_3 = 0 \Rightarrow x_3 = 0$, $x_1 = x_2 \Rightarrow X_3 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$.

Modal matrix $B = (X_1 \ X_2 \ X_3)$:

$$B = \begin{pmatrix} 1 & 19 & 1 \\ 0 & -6 & 1 \\ 0 & 15 & 0 \end{pmatrix}, \quad B^{-1} = \frac{\text{adj}(B)}{|B|}.$$

Now $|B| = 1(0 - 15) + 0 + 0 \Rightarrow |B| = -15$.

$$\text{adj}(B) = (\text{cofactor matrix of } B)^T = \begin{pmatrix} +(-15) & -(0) & +(0) \\ -(-15) & +(0) & -(15) \\ +(19+6) & -(1) & +(-6) \end{pmatrix}^T = \begin{pmatrix} -15 & 0 & 0 \\ 15 & 0 & -15 \\ 25 & -1 & -6 \end{pmatrix}^T$$

$$\Rightarrow \text{adj}(B) = \begin{pmatrix} -15 & 15 & 25 \\ 0 & 0 & -1 \\ 0 & -15 & -6 \end{pmatrix}.$$

$$\Rightarrow B^{-1} = \frac{\text{adj}(B)}{|B|} = \begin{pmatrix} 1 & -1 & -5/3 \\ 0 & 0 & 1/15 \\ 0 & 1 & 2/5 \end{pmatrix}.$$

Then

$$\begin{aligned} B^{-1}AB &= \begin{pmatrix} 1 & -1 & -5/3 \\ 0 & 0 & 1/15 \\ 0 & 1 & 2/5 \end{pmatrix} \begin{pmatrix} 1 & 2 & -3 \\ 0 & 3 & 2 \\ 0 & 0 & -2 \end{pmatrix} \begin{pmatrix} 1 & 19 & 1 \\ 0 & -6 & 1 \\ 0 & 15 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 1 & -1 & -5/3 \\ 0 & 0 & 1/15 \\ 0 & 1 & 2/5 \end{pmatrix} \begin{pmatrix} 1+0+0 & 19-12-45 & 1+2+0 \\ 0+0+0 & 0-18+30 & 0+3+0 \\ 0+0+0 & 0+0-30 & 0+0+0 \end{pmatrix} \\ &= \begin{pmatrix} 1 & -1 & -5/3 \\ 0 & 0 & 1/15 \\ 0 & 1 & 2/5 \end{pmatrix} \begin{pmatrix} 1 & -38 & 3 \\ 0 & 12 & 3 \\ 0 & -30 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 3 \end{pmatrix} = D. \end{aligned}$$

Hence A and D are similar matrices. ■

Note

- (1) Diagonalization of A is possible only if the eigen vectors of A are linearly independent.
- (2) We know that if the eigen values of A are distinct, then the corresponding eigen vectors are linearly independent; in that case diagonalization is possible.
- (3) If some of the eigen values of A are equal, and if $AX = \lambda X$ yields a single equation, then diagonalization is possible; otherwise diagonalization is not possible.

3.2 Orthogonal transformation

This transformation is used to diagonalize a *real symmetric* matrix.

Result 3.2 (Steps in the orthogonal transformation)

(i) Identify the eigen values and eigen vectors of the given matrix A ; let them be $\lambda_1, \lambda_2, \lambda_3$ and X_1, X_2, X_3 .

(ii) Normalize each eigen vector; let them be $N(X_1), N(X_2), N(X_3)$, where for $X_1 = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$,

$$N(X_1) = \begin{pmatrix} x_1/\sqrt{x_1^2 + x_2^2 + x_3^2} \\ x_2/\sqrt{x_1^2 + x_2^2 + x_3^2} \\ x_3/\sqrt{x_1^2 + x_2^2 + x_3^2} \end{pmatrix}.$$

(iii) Construct the modal matrix $P = (N(X_1) \ N(X_2) \ N(X_3))$; this matrix P is obviously an orthogonal matrix $\Rightarrow P^{-1} = P^T$.

(iv) Form the product P^TAP ; it will be a diagonal matrix D , i.e. $P^TAP = D$, and the entries of D are the eigen values of A .

Problem 3.2

Diagonalize the following real symmetric matrix $A = \begin{pmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{pmatrix}$ using orthogonal transformation.

Solution. Given A is a real symmetric matrix. Already we know (Problem 2 of §2) the characteristic equation is $\lambda^3 - 7\lambda^2 + 36 = 0$, the eigen values are $\lambda = -2, 3, 6$ and the eigen vectors are $X_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$, $X_2 = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$, $X_3 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$. Then

$$N(X_1) = \begin{pmatrix} \frac{1}{\sqrt{1^2+0^2+(-1)^2}} \\ 0 \\ \frac{-1}{\sqrt{1^2+0^2+(-1)^2}} \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} \\ 0 \\ -1/\sqrt{2} \end{pmatrix}; \quad \text{similarly} \quad N(X_2) = \begin{pmatrix} 1/\sqrt{3} \\ -1/\sqrt{3} \\ 1/\sqrt{3} \end{pmatrix}, \quad N(X_3) = \begin{pmatrix} 1/\sqrt{6} \\ 2/\sqrt{6} \\ 1/\sqrt{6} \end{pmatrix}.$$

$\therefore$ Modal matrix $P = (N(X_1) \ N(X_2) \ N(X_3))$

$$\Rightarrow P = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{3} & 1/\sqrt{6} \\ 0 & -1/\sqrt{3} & 2/\sqrt{6} \\ -1/\sqrt{2} & 1/\sqrt{3} & 1/\sqrt{6} \end{pmatrix}$$

is an orthogonal matrix, i.e. $P^T = P^{-1}$. Now consider the product

$$\begin{aligned} P^T A P &= \begin{pmatrix} 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 1/\sqrt{3} & -1/\sqrt{3} & 1/\sqrt{3} \\ 1/\sqrt{6} & 2/\sqrt{6} & 1/\sqrt{6} \end{pmatrix} \begin{pmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{3} & 1/\sqrt{6} \\ 0 & -1/\sqrt{3} & 2/\sqrt{6} \\ -1/\sqrt{2} & 1/\sqrt{3} & 1/\sqrt{6} \end{pmatrix} \\ &= \begin{pmatrix} 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 1/\sqrt{3} & -1/\sqrt{3} & 1/\sqrt{3} \\ 1/\sqrt{6} & 2/\sqrt{6} & 1/\sqrt{6} \end{pmatrix} \begin{pmatrix} -2/\sqrt{2} & 3/\sqrt{3} & 6/\sqrt{6} \\ 0 & -3/\sqrt{3} & 12/\sqrt{6} \\ 2/\sqrt{2} & 3/\sqrt{3} & 6/\sqrt{6} \end{pmatrix} \\ &= \begin{pmatrix} -2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6 \end{pmatrix} = D. \end{aligned}$$

■

Problem 3.3

Diagonalize the matrix $A = \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$ using orthogonal transformation.

Solution. Given A is a real symmetric matrix. Already we know the char. eqn. is $\lambda^3 - 12\lambda^2 + (8+14+14)\lambda - 32 = 0$, i.e. $\lambda^3 - 12\lambda^2 + 36\lambda - 32 = 0$. By substitution $\lambda = 2$ is one root ($8 - 48 + 72 - 32 = 0$); remaining roots are $\lambda = 2, 8 \Rightarrow$ eigen values are $\boxed{\lambda = 2, 2, 8}$.

Eigen vectors. (i) For $\lambda = 8$: $AX = \lambda X \Rightarrow -2x_1 - 2x_2 + 2x_3 = 0$ (1), $-2x_1 - 5x_2 - x_3 = 0$ (2), $2x_1 - x_2 - 5x_3 = 0$ (3). All the above equations are distinct. $\therefore$ Apply CMR on (1) and (2):

$$\frac{x_1}{2+10} = \frac{-x_2}{2+4} = \frac{x_3}{10-4} \Rightarrow \frac{x_1}{12} = \frac{x_2}{-6} = \frac{x_3}{6} \Rightarrow \frac{x_1}{2} = \frac{x_2}{-1} = \frac{x_3}{1} = k.$$

Take $k = 1$: eigen vector is $X_1 = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$.

(ii) For $\lambda = 2$: $AX = \lambda X \Rightarrow 4x_1 - 2x_2 + 2x_3 = 0, -2x_1 + x_2 - x_3 = 0, 2x_1 - x_2 + x_3 = 0$. All the above three equations are the same, i.e. it will be $\boxed{2x_1 - x_2 + x_3 = 0}$. Assume $x_3 = 0$: $2x_1 - x_2 = 0 \Rightarrow 2x_1 = x_2 \Rightarrow \frac{x_1}{1} = \frac{x_2}{2} = k \Rightarrow$ eigen vector $X_2 = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$.

(iii) Again for $\lambda = 2$, let it be $X_3 = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$. We know that the eigen vectors of a real symmetric matrix are *pairwise orthogonal* ($X_1^T X_2 = 0, X_2^T X_3 = 0, X_3^T X_1 = 0$). $\therefore$ We must have $X_3^T X_1 = 0$ and $X_3^T X_2 = 0$:

$$\Rightarrow (a \ b \ c) \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} = 0 \text{ and } (a \ b \ c) \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} = 0 \Rightarrow 2a - b + c = 0, \quad a + 2b + 0c = 0.$$

Using CMR,

$$\frac{a}{0-2} = \frac{-b}{0-1} = \frac{c}{4+1} \Rightarrow \frac{a}{-2} = \frac{b}{1} = \frac{c}{5} \Rightarrow X_3 = \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \\ 5 \end{pmatrix}.$$

Now

$$N(X_1) = \begin{pmatrix} 2/\sqrt{6} \\ -1/\sqrt{6} \\ 1/\sqrt{6} \end{pmatrix}, \quad N(X_2) = \begin{pmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \\ 0 \end{pmatrix}, \quad N(X_3) = \begin{pmatrix} -2/\sqrt{30} \\ 1/\sqrt{30} \\ 5/\sqrt{30} \end{pmatrix}.$$

$\Rightarrow$ Modal matrix $P = (N(X_1) \ N(X_2) \ N(X_3))$,

$$P = \begin{pmatrix} 2/\sqrt{6} & 1/\sqrt{5} & -2/\sqrt{30} \\ -1/\sqrt{6} & 2/\sqrt{5} & 1/\sqrt{30} \\ 1/\sqrt{6} & 0 & 5/\sqrt{30} \end{pmatrix}.$$

Then

$$\begin{aligned} P^T A P &= \begin{pmatrix} 2/\sqrt{6} & -1/\sqrt{6} & 1/\sqrt{6} \\ 1/\sqrt{5} & 2/\sqrt{5} & 0 \\ -2/\sqrt{30} & 1/\sqrt{30} & 5/\sqrt{30} \end{pmatrix} \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix} P \\ &= \begin{pmatrix} 2/\sqrt{6} & -1/\sqrt{6} & 1/\sqrt{6} \\ 1/\sqrt{5} & 2/\sqrt{5} & 0 \\ -2/\sqrt{30} & 1/\sqrt{30} & 5/\sqrt{30} \end{pmatrix} \begin{pmatrix} 16/\sqrt{6} & 2/\sqrt{5} & -4/\sqrt{30} \\ -8/\sqrt{6} & 4/\sqrt{5} & 2/\sqrt{30} \\ 8/\sqrt{6} & 0 & 10/\sqrt{30} \end{pmatrix} \\ &= \begin{pmatrix} 8 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} = D. \end{aligned}$$

■

Problem 3.4

Diagonalize the matrix $A = \begin{pmatrix} 2 & -2 & 2 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{pmatrix}$ if possible.

Solution. From the earlier problem, the eigen values are $\lambda = 2, 2, -2$, and for the repeated eigen value $\lambda = 2$ the equations from $AX = \lambda X$ are *distinct* (they do not reduce to a single equation). Hence we cannot find two independent eigen vectors for $\lambda = 2$ ($X_3 = X_2$), the eigen vectors of A are not linearly independent, and therefore diagonalization of A is **not possible**. ■

4 Applications of diagonalization: quadratic forms

4.1 Conversion of a quadratic form to canonical form

Any 3-D quadratic surface can be represented in the following general form:

$$ax^2 + by^2 + cz^2 + 2hxy + 2fyz + 2gzx = k. \quad (1)$$

This represents the general form for quadratic surfaces such as ellipsoid, paraboloid, etc. Equation (1) is called the **quadratic form** of any quadratic surface. We can represent (1) in the form of a matrix as follows:

$$(x \ y \ z) \begin{pmatrix} a & h & g \\ h & b & f \\ g & f & c \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = k, \quad \text{i.e. } X^T A X = k, \quad \text{where } X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

and $A =$ coefficient matrix $= \begin{pmatrix} a & h & g \\ h & b & f \\ g & f & c \end{pmatrix}$. This coefficient matrix A is a real symmetric matrix.

Result 4.1 (Coefficient matrix of a quadratic form)

The elements of the coefficient matrix A are obtained from the given quadratic form as follows:

$$\begin{array}{l|l} a_{11} = \text{coeff. of } x^2 & a_{12} = a_{21} = \frac{1}{2}(\text{coeff. of } xy) \\ a_{22} = \text{coeff. of } y^2 & a_{13} = a_{31} = \frac{1}{2}(\text{coeff. of } xz) \\ a_{33} = \text{coeff. of } z^2 & a_{23} = a_{32} = \frac{1}{2}(\text{coeff. of } yz) \end{array}$$

Since A is a real symmetric matrix, if we diagonalize A through the orthogonal transformation we get a diagonal matrix D which is similar to A . Through this diagonal matrix D , if we form a quadratic surface equation or quadratic form, it will be

$$Y^T D Y = k \Rightarrow (x' \ y' \ z') \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix} \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = k \Rightarrow \boxed{\lambda_1 x'^2 + \lambda_2 y'^2 + \lambda_3 z'^2 = k}$$

Definition 4.1 (Canonical form)

The form $\lambda_1 x'^2 + \lambda_2 y'^2 + \lambda_3 z'^2 = k$ (with λ_i the eigen values of the coefficient matrix A) is called the **canonical form** of the given quadratic form.

This canonical form represents the given quadratic surface when it is rotated through an angle θ ; based on its new principal axes x', y', z' we have the equation of the quadratic surface. The rotation angle θ is obtained from the first element of the modal matrix P , i.e.

$$\boxed{\theta = \cos^{-1}(p_{11})}, \quad \text{where } p_{11} = \text{first element of } P.$$

Problem 4.1

Perform a rotation of axes to eliminate the xy , yz and zx terms in the quadratic equation $x^2 + 5y^2 + z^2 + 2xy + 6xz + 2yz = 0$, i.e. identify the canonical form of the given quadratic form.

Solution. The given quadratic form is $x^2 + 5y^2 + z^2 + 2xy + 6xz + 2yz = 0$. The matrix form of the above quadratic surface is

$$(x \ y \ z) \begin{pmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0, \quad \text{i.e. } X^T A X = 0, \quad \text{where } A = \begin{pmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{pmatrix}$$

is a real symmetric matrix. To eliminate the xy , yz and zx terms of the given quadratic form, we have to perform diagonalization for A . We know already the eigen values of A are $\lambda = -2, 3, 6$ and the eigen vectors are

$X_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$, $X_2 = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$ and $X_3 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$. Their normalized forms are

$$N(X_1) = \begin{pmatrix} 1/\sqrt{2} \\ 0 \\ -1/\sqrt{2} \end{pmatrix}, \quad N(X_2) = \begin{pmatrix} 1/\sqrt{3} \\ -1/\sqrt{3} \\ 1/\sqrt{3} \end{pmatrix}, \quad N(X_3) = \begin{pmatrix} 1/\sqrt{6} \\ 2/\sqrt{6} \\ 1/\sqrt{6} \end{pmatrix}$$

$$\Rightarrow \text{Modal matrix } P = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{3} & 1/\sqrt{6} \\ 0 & -1/\sqrt{3} & 2/\sqrt{6} \\ -1/\sqrt{2} & 1/\sqrt{3} & 1/\sqrt{6} \end{pmatrix}.$$

Then $P^T A P = D = \begin{pmatrix} -2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6 \end{pmatrix}$. Hence the canonical form of the given quadratic form is $Y^T D Y = 0$:

$$\Rightarrow (x' \ y' \ z') \begin{pmatrix} -2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6 \end{pmatrix} \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = 0 \Rightarrow -2x'^2 + 3y'^2 + 6z'^2 = 0,$$

or generally, $-2x^2 + 3y^2 + 6z^2 = 0$ is the canonical form of the given quadratic form.

Here the rotation angle $\theta = \cos^{-1}(\text{1st element of } P) = \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) \Rightarrow \boxed{\theta = 45^\circ}$, i.e. under $\theta = 45^\circ$ of rotation we get the quadratic form without xy , yz and zx terms. ■

Problem 4.2

Perform a rotation of axes to eliminate the xz term in the quadratic equation $5x^2 + 4y^2 + 5z^2 + 8xz = 36$.

Solution. The given Q.F. is $5x^2 + 4y^2 + 5z^2 + 8xz = 36$. Here the matrix representation is

$$(x \ y \ z) \begin{pmatrix} 5 & 0 & 4 \\ 0 & 4 & 0 \\ 4 & 0 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 36, \quad A = \begin{pmatrix} 5 & 0 & 4 \\ 0 & 4 & 0 \\ 4 & 0 & 5 \end{pmatrix} \text{ is a real symmetric matrix.}$$

Characteristic equation: $\lambda^3 - 14\lambda^2 + (20+9+20)\lambda - 36 = 0$, i.e. $\lambda^3 - 14\lambda^2 + 49\lambda - 36 = 0 \Rightarrow (\lambda-1)(\lambda-4)(\lambda-9) = 0$.
 $\Rightarrow$ Eigen values are $\lambda = 1, 4, 9$, and the eigen vectors are

$$X_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \quad X_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad X_3 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}.$$

(For $\lambda = 1$: $4x_1 + 4x_3 = 0$, $3x_2 = 0$; for $\lambda = 4$: $x_1 + 4x_3 = 0$, $4x_1 + x_3 = 0 \Rightarrow x_1 = x_3 = 0$; for $\lambda = 9$: $-4x_1 + 4x_3 = 0$, $-5x_2 = 0$.) Normalizing,

$$P = \begin{pmatrix} 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 0 & 1 & 0 \\ -1/\sqrt{2} & 0 & 1/\sqrt{2} \end{pmatrix}, \quad P^T A P = D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 9 \end{pmatrix},$$

and the rotation angle $\theta = \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) \Rightarrow \boxed{\theta = 45^\circ}$.

Then the canonical form is $Y^T D Y = 36$:

$$\Rightarrow x'^2 + 4y'^2 + 9z'^2 = 36 \quad \text{generally} \quad x^2 + 4y^2 + 9z^2 = 36 \Rightarrow \boxed{\frac{x^2}{6^2} + \frac{y^2}{3^2} + \frac{z^2}{2^2} = 1}$$

is the ellipsoid equation. ■

4.2 Nature of a quadratic form (Q.F.)

Definition 4.2 (Nature of a Q.F. via eigen values)

The nature of a Q.F. $X^T A X$ can be classified in the following ways:

- (1) **Positive definite:** if all the eigen values of A are positive (> 0).
- (2) **Positive semi-definite:** if the eigen values of A are ≥ 0 and at least one eigen value is zero.
- (3) **Negative definite:** if all the eigen values of A are < 0 .
- (4) **Negative semi-definite:** if the eigen values of A are ≤ 0 and at least one eigen value is zero.
- (5) **Indefinite:** if the eigen values of A are both positive and negative.

Another way to find the nature of the Q.F.

In this method, without finding the eigen values of A and only by the determinant values of sub-matrices of A , we can determine the nature of the Q.F. For

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}, \quad D_1 = |a_{11}|, \quad D_2 = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}, \quad D_3 = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}, \quad \dots, \quad D_n = |A|.$$

Result 4.2 (Nature of a Q.F. via principal minors)

- (1) If all the values of D_i ($i = 1$ to n) are > 0 , then the Q.F. is said to be **positive definite**.
- (2) If all the values of D_i are ≥ 0 and at least one $D_i = 0$, then the Q.F. is said to be **positive semi-definite**.
- (3) If $D_1, D_3, D_5, \dots < 0$ and $D_2, D_4, D_6, \dots > 0$, i.e. $(-1)^n D_n > 0$, then the Q.F. is said to be **negative definite**.
- (4) If $D_1, D_3, D_5, \dots \leq 0$ and $D_2, D_4, D_6, \dots \geq 0$, i.e. $(-1)^n D_n \geq 0$, and at least one $D_i = 0$, then the Q.F. is said to be **negative semi-definite**.
- (5) If all the above conditions fail, then the Q.F. is said to be **indefinite**.

4.3 Rank, index and signature of the Q.F.**Definition 4.3 (Rank, index, signature)**

Rank: the number of terms in the canonical form of the given Q.F. is called the rank of the Q.F.

Index: the number of positive terms in the canonical form of the given Q.F. is called the index of the Q.F.

Signature: the difference between the number of positive terms and the number of non-positive terms in the canonical form is called the signature of the Q.F.

Problem 4.3

Determine the nature, rank, index and signature of the Q.F. $x^2 + 5y^2 + z^2 + 2xy + 6xz + 2yz = 0$.

Solution. The given Q.F. can be written as

$$(x \ y \ z) \begin{pmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \Rightarrow A = \begin{pmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{pmatrix},$$

and we know the eigen values of A are $\lambda = -2, 3, 6$; the respective canonical form is $-2x^2 + 3y^2 + 6z^2 = 0$. Here the eigen values of A are both positive and negative $\Rightarrow$ the nature of the Q.F. is **indefinite**.

Also from the canonical form $-2x^2 + 3y^2 + 6z^2 = 0$, we see that

$$\text{Rank} = 3, \quad \text{Index} = 2, \quad \text{Signature} = 2 - 1 = 1.$$

Another way. With $A = \begin{pmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{pmatrix}$:

$$D_1 = |1| = 1, \quad D_2 = \begin{vmatrix} 1 & 1 \\ 1 & 5 \end{vmatrix} = 4, \quad D_3 = \begin{vmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{vmatrix} = 1(4) - 1(-2) + 3(-14) = 4 + 2 - 42 \Rightarrow D_3 = -36.$$

$D_1 > 0$, $D_2 > 0$ and $D_3 < 0 \Rightarrow$ indefinite in nature. ■

Problem 4.4

Determine the nature of the Q.F. $2x_1^2 + x_2^2 + x_3^2 + 2x_1x_2 - 2x_1x_3 - 4x_2x_3 = 0$ and also find its rank, index and signature.

Solution.

$$A = \begin{pmatrix} 2 & 1 & -1 \\ 1 & 1 & -2 \\ -1 & -2 & 1 \end{pmatrix}.$$

Characteristic equation: $\lambda^3 - 4\lambda^2 + (-3+1+1)\lambda - |A| = 0$, where $|A| = 2(1-4) - 1(1-2) - 1(-2+1) = -6+1+1 = -4$, i.e. $\lambda^3 - 4\lambda^2 - \lambda + 4 = 0 \Rightarrow (\lambda - 1)(\lambda + 1)(\lambda - 4) = 0$. The eigen values of A are $\lambda = -1, 1, 4 \Rightarrow$ **indefinite**. The canonical form is $-y_1^2 + y_2^2 + 4y_3^2 = 0$, so

$$\text{Rank} = 3, \quad \text{Index} = 2, \quad \text{Signature} = 1.$$

(Check: $D_1 = 2 > 0$, $D_2 = 1 > 0$, $D_3 = -4 < 0 \Rightarrow$ indefinite.) ■

Exercise. Reduce the Q.F. $x_1^2 + 2x_2^2 + x_3^2 - 2x_1x_2 + 2x_2x_3 = 0$ to the canonical form. Also find the rank, index, signature and nature of the Q.F.

(Hint: $\lambda = 1, 0, 3 \Rightarrow$ positive semi-definite.)

4.4 Power of a matrix using diagonalization

Using the similarity transformation, we have $D = B^{-1}AB$, where A is the given matrix. Then

$$BDB^{-1} = BB^{-1}ABB^{-1} = A \Rightarrow \boxed{A = BDB^{-1}}$$

$$A^2 = A \cdot A = (BDB^{-1})(BDB^{-1}) = BD(B^{-1}B)DB^{-1} \Rightarrow \boxed{A^2 = BD^2B^{-1}}$$

Result 4.3 (Power of a matrix)

Similarly $A^3 = BD^3B^{-1}$, and in general

$$A^n = BD^nB^{-1}, \quad \text{where } D^n = \text{diag}(\lambda_1^n, \lambda_2^n, \lambda_3^n).$$

5 Properties of eigen values

Result 5.1 (Property 1)

- (i) Sum of the eigen values of $A =$ Sum of the diagonal elements of A .
- (ii) Product of the eigen values of $A = |A|$.

(Already proved.) We will see some more properties further.

Result 5.2 (Property 2)

Both A and A^T have the same eigen values.

Proof. For A , the characteristic equation is $|A - \lambda I| = 0$. We also know $|A| = |A^T|$, so

$$|A - \lambda I| = |(A - \lambda I)^T| = |A^T - \lambda I^T| = |A^T - \lambda I|,$$

i.e. $|A - \lambda I| = |A^T - \lambda I| \Rightarrow$ the characteristic polynomials of both A and A^T are the same $\Rightarrow$ their characteristic equations are the same $\Rightarrow$ they (A and A^T) have the same eigen values.

Result 5.3 (Property 3)

If λ is an eigen value of A , then λ^2 is an eigen value of A^2 . More generally, λ^k is an eigen value of A^k , where k is a positive integer.

Proof. Given λ is an eigen value of A :

$$\Rightarrow AX = \lambda X \quad (1)$$

Pre-multiplying by A on both sides: $A(AX) = A(\lambda X)$

$$\Rightarrow A^2X = \lambda(AX) = \lambda(\lambda X) \quad (\because \text{by (1)})$$

$$\Rightarrow A^2X = \lambda^2X \Rightarrow \lambda^2 \text{ is an eigen value of } A^2.$$

Result 5.4 (Property 4)

If λ is an eigen value of A , then $\frac{1}{\lambda}$ is an eigen value of A^{-1} .

Proof. Given λ is an eigen value of $A \Rightarrow AX = \lambda X$. Now pre-multiply by A^{-1} on both sides:

$$A^{-1}(AX) = A^{-1}(\lambda X) \Rightarrow (A^{-1}A)X = \lambda(A^{-1}X) \Rightarrow IX = \lambda A^{-1}X$$

$$\Rightarrow X = \lambda A^{-1}X \Rightarrow A^{-1}X = \left(\frac{1}{\lambda}\right)X \Rightarrow \boxed{\frac{1}{\lambda} \text{ is an eigen value of } A^{-1}}$$

Result 5.5 (Property 5)

If λ is an eigen value of A , then $k\lambda$ is an eigen value of kA , where k is a constant.

Proof. Given λ is an eigen value of A : $AX = \lambda X \Rightarrow kAX = k\lambda X \Rightarrow (kA)X = (k\lambda)X \Rightarrow k\lambda$ is an eigen value of kA .

Note

Remark. For upper and lower triangular matrices (such as $\begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{pmatrix}$ and $\begin{pmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 4 & 5 & 6 \end{pmatrix}$), the diagonal entries are the eigen values ($|A - \lambda I| = 0$ is just the product of the diagonal terms). $\therefore$ For the above upper triangular matrix the eigen values are 1, 4, 6; similarly for the lower triangular matrix the eigen values are 1, 3, 6.

5.1 Problems

Problem 5.1

Find the eigen values of $2A^2$, if $A = \begin{pmatrix} 4 & 1 \\ 3 & 2 \end{pmatrix}$.

Solution. The char. eqn. of A is $\lambda^2 - 6\lambda + (5) = 0 \Rightarrow (\lambda - 1)(\lambda - 5) = 0 \Rightarrow \lambda = 1, 5$. $\Rightarrow$ Eigen values of A are $\lambda = 1, 5 \Rightarrow$ eigen values of A^2 are $\lambda^2 = 1, 25 \Rightarrow$ eigen values of $2A^2$ are $2\lambda^2 = 2, 50$. ■

Problem 5.2

Prove that the eigen values of $-3A^{-1}$ are the same as those of $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$.

Solution. The char. eqn. is $\lambda^2 - 2\lambda - 3 = 0 \Rightarrow (\lambda - 3)(\lambda + 1) = 0 \Rightarrow$ eigen values of A are $\lambda = -1, 3$. $\Rightarrow$ Eigen values of A^{-1} are $\frac{1}{\lambda} = -1, \frac{1}{3} \Rightarrow$ eigen values of $-3A^{-1}$ are $-\frac{3}{\lambda} = 3, -1$, which are the same as those of A . ■

Problem 5.3

Find the sum of the squares of the eigen values of $A = \begin{pmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{pmatrix}$.

Solution. Since A is an upper triangular matrix, the eigen values of A are 3, 2, 5. $\Rightarrow$ Sum of the squares of the eigen values $= 3^2 + 2^2 + 5^2 = 9 + 4 + 25 = 38$. ■

Problem 5.4

If the sum of two eigen values and the trace of a 3×3 matrix are equal, then find $|A|$.

Solution. From the given condition, $\lambda_1 + \lambda_2 = \lambda_1 + \lambda_2 + \lambda_3$ (trace of A = sum of the diagonal elements = sum of the eigen values) $\Rightarrow \boxed{\lambda_3 = 0}$. Hence the determinant value of A is $|A| = \lambda_1 \lambda_2 \lambda_3 = 0 \Rightarrow A$ is a singular matrix. ■

Problem 5.5

The product of two eigen values of the matrix $A = \begin{pmatrix} 6 & -2 & 2 \\ -3 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$ is 16. Find the third eigen value.

Solution. Let λ_1, λ_2 and λ_3 be the three eigen values of A . Given $\lambda_1 \lambda_2 = 16$. But we know $\lambda_1 \lambda_2 \lambda_3 = |A|$ (product of the eigen values). Here $|A| = 6(9 - 1) + 2(-9 + 2) + 2(3 - 6) = 48 - 14 - 6 = 28$.

$$\Rightarrow 16\lambda_3 = |A| = 28 \Rightarrow \lambda_3 = \frac{28}{16} = \frac{7}{4}.$$

Problem 5.6

Find the eigen values of the matrix $A = \begin{pmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{pmatrix}$ and what can you infer about the matrix A ?

Solution. The char. eqn. is $\lambda^3 - 18\lambda^2 + 45\lambda = 0 \Rightarrow \lambda(\lambda - 3)(\lambda - 15) = 0$; the eigen values of A are $\lambda = 0, 3, 15$. We know the product of eigen values $= |A| \Rightarrow |A| = 0$ (since $\lambda = 0$) $\Rightarrow$ the given matrix A is a **singular** matrix. ■

Problem 5.7

Verify that the matrix $A = \frac{1}{3} \begin{pmatrix} -1 & 2 & -2 \\ -2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix}$ is orthogonal and that its eigen values are of unit modulus.

Solution.

$$AA^T = \frac{1}{3} \begin{pmatrix} -1 & 2 & -2 \\ -2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix} \cdot \frac{1}{3} \begin{pmatrix} -1 & -2 & 2 \\ 2 & 1 & 2 \\ -2 & 2 & 1 \end{pmatrix} = \frac{1}{9} \begin{pmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\Rightarrow AA^T = I \Rightarrow A \text{ is an orthogonal matrix.}$$

Then consider the matrix $3A = \begin{pmatrix} -1 & 2 & -2 \\ -2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix}$. Its char. eqn. is $\lambda^3 - \lambda^2 + 3\lambda - 27 = 0$, i.e. $(\lambda - 3)(\lambda^2 + 2\lambda + 9) = 0$,

so the eigen values of $3A$ are $3, -1 \pm i2\sqrt{2}$. $\Rightarrow$ Eigen values of A are $1, \frac{-1 + i2\sqrt{2}}{3}, \frac{-1 - i2\sqrt{2}}{3}$. Now $|1| = 1$ and

$$\left| \frac{-1 \pm i2\sqrt{2}}{3} \right| = \frac{1}{3} \sqrt{(-1)^2 + (2\sqrt{2})^2} = \frac{1}{3} \sqrt{1 + 8} = 1.$$

$\Rightarrow$ The eigen values are of unit modulus. ■

6 Solution of systems of equations

Consider a system of m equations with n unknowns such as

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m. \end{aligned}$$

We can write the above system of equations in matrix form such as

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}, \quad \text{i.e. } AX = B,$$

where A is the coefficient matrix of order $m \times n$, X is the unknown column matrix and B is the RHS elements column matrix.

Consider the **augmented matrix**

$$(A, B) = \left(\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & & & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array} \right).$$

Now apply elementary row transformations and convert the matrix (A, B) to a row-echelon form matrix. We can determine the solution of the given system from the row-echelon form as follows. Suppose we have the row-echelon form of (A, B) as

$$(A, B) \sim \left(\begin{array}{ccccc|c} d_1 & k_1 & k_2 & \cdots & k_{n-1} & c_1 \\ 0 & d_2 & l_1 & \cdots & l_{n-2} & c_2 \\ 0 & 0 & d_3 & \cdots & m_{n-3} & c_3 \\ \vdots & & & \ddots & & \vdots \\ 0 & 0 & 0 & \cdots & g_n & c_m \end{array} \right).$$

Result 6.1 (Consistency of $AX = B$)

- (i) If $R(A, B) \neq R(A)$, then the given system of equations is **inconsistent** and the system has no solution.
- (ii) If $R(A, B) = R(A) = n$, the number of unknowns, then the given system of equations is **consistent and has a unique solution**.
- (iii) If $R(A, B) = R(A) < n$, the number of unknowns, then the given system of equations is **consistent but has infinitely many solutions**.

Problem 6.1

Solve the following system of equations if it is consistent: $x + y + z = 3$; $2x + y - z = 2$ and $4x - y + 2z = 5$.

Solution. The matrix form of the above system of equations is

$$\begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ 4 & -1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ 5 \end{pmatrix}, \quad \text{i.e. of the form } AX = B.$$

Then the augmented matrix is

$$(A, B) = \left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 2 & 1 & -1 & 2 \\ 4 & -1 & 2 & 5 \end{array} \right).$$

Apply $R_2 \rightarrow R_2 - 2R_1$, $R_3 \rightarrow R_3 - 4R_1$:

$$(A, B) \sim \left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & -1 & -3 & -4 \\ 0 & -5 & -2 & -7 \end{array} \right).$$

Apply $R_3 \rightarrow R_3 - 5R_2$ ($-2 + 15 = 13$, $-7 + 20 = 13$):

$$(A, B) \sim \left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & -1 & -3 & -4 \\ 0 & 0 & 13 & 13 \end{array} \right),$$

which is the row-echelon form of the matrix (A, B) . In this row-echelon form, $R(A, B) = 3$ and $R(A) = 3$, i.e. $R(A, B) = R(A) = 3 =$ the number of unknowns. $\therefore$ The given system of equations is consistent and has a unique solution.

To get the solution, write the row-echelon form matrix (A, B) as $AX = B$ form, i.e.

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & -3 \\ 0 & 0 & 13 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3 \\ -4 \\ 13 \end{pmatrix} \Rightarrow x + y + z = 3 \quad (1), \quad -y - 3z = -4 \quad (2), \quad 13z = 13 \quad (3).$$

From (3) we get $\boxed{z = 1}$; substituting in (2), $\boxed{y = 1}$; substituting these in (1), $\boxed{x = 1}$ is the required solution. ■

Problem 6.2

Solve the following if they are consistent: $2x - 3y + 7z = 5$; $3x + y - 3z = 13$ and $2x + 19y - 47z = 32$.

Solution. The given system of equations can be written in $AX = B$ form:

$$\begin{pmatrix} 2 & -3 & 7 \\ 3 & 1 & -3 \\ 2 & 19 & -47 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 5 \\ 13 \\ 32 \end{pmatrix}. \quad \text{Then } (A, B) = \left(\begin{array}{ccc|c} 2 & -3 & 7 & 5 \\ 3 & 1 & -3 & 13 \\ 2 & 19 & -47 & 32 \end{array} \right).$$

Apply $R_2 \rightarrow 2R_2 - 3R_1$, $R_3 \rightarrow R_3 - R_1$ ($2 + 9 = 11$, $-6 - 21 = -27$, $26 - 15 = 11$):

$$(A, B) \sim \left(\begin{array}{ccc|c} 2 & -3 & 7 & 5 \\ 0 & 11 & -27 & 11 \\ 0 & 22 & -54 & 27 \end{array} \right).$$

Apply $R_3 \rightarrow R_3 - 2R_2$ ($27 - 22 = 5$):

$$(A, B) \sim \left(\begin{array}{ccc|c} 2 & -3 & 7 & 5 \\ 0 & 11 & -27 & 11 \\ 0 & 0 & 0 & 5 \end{array} \right),$$

which is the row-echelon form. For the given (A, B) we have $R(A, B) = 3$, $R(A) = 2 \Rightarrow R(A, B) \neq R(A) \Rightarrow$ the given system is **inconsistent** and has no solution. ■

Problem 6.3

Solve the following system of equations if they are consistent: $x + 2y + 2z = 1$; $2x + y + z = 2$; $3x + 2y + 2z = 3$ and $y + z = 0$.

Solution.

$$(A, B) = \left(\begin{array}{ccc|c} 1 & 2 & 2 & 1 \\ 2 & 1 & 1 & 2 \\ 3 & 2 & 2 & 3 \\ 0 & 1 & 1 & 0 \end{array} \right).$$

Apply $R_2 \rightarrow R_2 - 2R_1$, $R_3 \rightarrow R_3 - 3R_1$:

$$(A, B) \sim \left(\begin{array}{ccc|c} 1 & 2 & 2 & 1 \\ 0 & -3 & -3 & 0 \\ 0 & -4 & -4 & 0 \\ 0 & 1 & 1 & 0 \end{array} \right). \quad \text{Apply } R_4 \leftrightarrow R_2: \quad (A, B) \sim \left(\begin{array}{ccc|c} 1 & 2 & 2 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & -4 & -4 & 0 \\ 0 & -3 & -3 & 0 \end{array} \right).$$

Apply $R_3 \rightarrow R_3 + 4R_2$, $R_4 \rightarrow R_4 + 3R_2$:

$$(A, B) \sim \left(\begin{array}{ccc|c} 1 & 2 & 2 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right),$$

which is the row-echelon form of the matrix (A, B) . In this form $R(A, B) = 2$; $R(A) = 2$, i.e. $R(A, B) = R(A) = 2 < 3$, the number of unknowns. $\therefore$ The given system is consistent but has many solutions.

To get the solution, write the row-echelon form of (A, B) as $AX = B$:

$$x + 2y + 2z = 1 \quad (1), \quad y + z = 0 \quad (2).$$

From (2), $y = -z$; if we take $\boxed{z = k}$, $\Rightarrow \boxed{y = -k}$. By (1), $x = 1 - 2z - 2y = 1 - 2k + 2k \Rightarrow \boxed{x = 1}$. $\therefore$ The solution is $x = 1$; $y = -k$; $z = k$, where k takes any value. ■

Problem 6.4

For what value of λ do the equations $x + y + z = 1$, $x + 2y + 4z = \lambda$, $x + 4y + 10z = \lambda^2$ have a solution? Solve completely in each case.

Solution. The matrix form of the given system of equations will be

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 4 & 10 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ \lambda \\ \lambda^2 \end{pmatrix} \text{ of the form } AX = B. \quad \text{Then } (A, B) = \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & \lambda \\ 1 & 4 & 10 & \lambda^2 \end{array} \right).$$

Apply $R_2 \rightarrow R_2 - R_1$, $R_3 \rightarrow R_3 - R_1$:

$$(A, B) \sim \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & \lambda - 1 \\ 0 & 3 & 9 & \lambda^2 - 1 \end{array} \right).$$

Apply $R_3 \rightarrow R_3 - 3R_2$ ($\lambda^2 - 1 - 3(\lambda - 1) = \lambda^2 - 3\lambda + 2$):

$$(A, B) \sim \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & \lambda - 1 \\ 0 & 0 & 0 & \lambda^2 - 3\lambda + 2 \end{array} \right),$$

which is the row-echelon form of the given matrix (A, B) .

Suppose $\lambda^2 - 3\lambda + 2 \neq 0$; then $R(A, B) = 3$, $R(A) = 2 \Rightarrow$ the system is inconsistent and has no solution.

Suppose $\lambda^2 - 3\lambda + 2 = 0$; then $R(A, B) = 2 = R(A) <$ the number of unknowns. $\therefore$ The system is consistent and has infinitely many solutions, i.e. $\lambda^2 - 3\lambda + 2 = 0 \Rightarrow (\lambda - 1)(\lambda - 2) = 0 \Rightarrow \boxed{\lambda = 1, 2}$; at these values of λ , the system has many solutions.

To get the solution, write the row-echelon form of (A, B) as $AX = B$:

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ \lambda - 1 \\ \lambda^2 - 3\lambda + 2 \end{pmatrix} \Rightarrow x + y + z = 1 \quad (1), \quad y + 3z = \lambda - 1 \quad (2).$$

Case (i): when $\lambda = 1$, (1) and (2) $\Rightarrow x + y + z = 1$, $y + 3z = 0$. $y = -3z$; take $z = k \Rightarrow \boxed{y = -3k}$, and $x = 1 - y - z = 1 + 3k - k \Rightarrow \boxed{x = 1 + 2k}$. $\therefore$ When $\lambda = 1$, the solution is $x = 1 + 2k$, $y = -3k$ and $z = k$, and k takes any value.

Case (ii): when $\lambda = 2$, (1) and (2) $\Rightarrow x + y + z = 1$, $y + 3z = 1$. (2) $\Rightarrow y = 1 - 3z$; when $z = k$, $\boxed{y = 1 - 3k}$. (1) $\Rightarrow x = 1 - (1 - 3k) - k \Rightarrow \boxed{x = 2k}$. $\therefore$ When $\lambda = 2$, the solution is $x = 2k$, $z = k$ and $y = 1 - 3k$, and k takes any value. ■

Problem 6.5

For what values of λ and μ does the system of equations $x + 2y + z = 8$; $2x + y + 3z = 13$; $3x + 4y - \lambda z = \mu$ have (i) no solution, (ii) a unique solution, (iii) infinitely many solutions?

Solution. The augmented matrix of the given system of equations is

$$(A, B) = \left(\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 2 & 1 & 3 & 13 \\ 3 & 4 & -\lambda & \mu \end{array} \right).$$

Apply $R_2 \rightarrow R_2 - 2R_1$, $R_3 \rightarrow R_3 - 3R_1$:

$$(A, B) \sim \left(\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 0 & -3 & 1 & -3 \\ 0 & -2 & -\lambda - 3 & \mu - 24 \end{array} \right). \quad \text{Apply } R_2 \rightarrow -\frac{1}{3}R_2: \quad (A, B) \sim \left(\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 0 & 1 & -\frac{1}{3} & 1 \\ 0 & -2 & -\lambda - 3 & \mu - 24 \end{array} \right).$$

Apply $R_3 \rightarrow R_3 + 2R_2$ ($-\lambda - 3 - \frac{2}{3} = -\lambda - \frac{11}{3}$, $\mu - 24 + 2 = \mu - 22$):

$$(A, B) \sim \left(\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 0 & 1 & -\frac{1}{3} & 1 \\ 0 & 0 & -\lambda - \frac{11}{3} & \mu - 22 \end{array} \right),$$

which is the row-echelon form of (A, B) . From the above (A, B) we see that:

(i) The system possesses no solution only if $-\lambda - \frac{11}{3} = 0$ and $\mu - 22 \neq 0$, i.e.

$$\boxed{\text{no solution only if } \lambda = -\frac{11}{3} \text{ and } \mu \neq 22}$$

(ii) The system possesses a unique solution if $-\lambda - \frac{11}{3} \neq 0$, and $\mu - 22$ may be any value, i.e.

$$\boxed{\text{unique solution only if } \lambda \neq -\frac{11}{3} \text{ and } \mu \text{ takes any value}}$$

(iii) The system possesses infinitely many solutions only if $-\lambda - \frac{11}{3} = 0$ and $\mu - 22 = 0$, i.e.

$$\boxed{\text{infinitely many solutions if } \lambda = -\frac{11}{3} \text{ and } \mu = 22}$$

■

6.1 Solution of a system of homogeneous simultaneous equations

They must be in the form $AX = 0$, i.e.

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \Rightarrow \begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = 0 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = 0. \end{cases}$$

Here $(A, B) = \left(\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & 0 \\ \vdots & & & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & 0 \end{array} \right)$, and after the row-echelon form of (A, B) , always $R(A, B) = R(A)$.

Result 6.2 (Homogeneous systems)

- (i) If $R(A) = R(A, B) = n$, the number of unknowns, the system has a unique solution, and it will be the **trivial solution**, i.e. $x_1 = 0, x_2 = 0, \dots, x_n = 0$.
- (ii) If $R(A) = R(A, B) < n$, the system will have infinitely many solutions (**non-trivial solutions**).

Problem 6.6

Find the non-trivial solution of the equations $x + 2y + 3z = 0$, $3x + 4y + 4z = 0$ and $7x + 10y + 11z = 0$, if it exists.

Solution.

$$(A, B) = \left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 3 & 4 & 4 & 0 \\ 7 & 10 & 11 & 0 \end{array} \right). \text{ Apply } R_2 \rightarrow R_2 - 3R_1, R_3 \rightarrow R_3 - 7R_1: (A, B) \sim \left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & -2 & -5 & 0 \\ 0 & -4 & -10 & 0 \end{array} \right).$$

Apply $R_3 \rightarrow R_3 - 2R_2$:

$$(A, B) \sim \left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & -2 & -5 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right),$$

which is the row-echelon form. Here $R(A, B) = R(A) = 2 < \text{the number of unknowns}$. $\therefore$ The system possesses infinitely many solutions. To get the solution, write (A, B) as $AX = B$:

$$x + 2y + 3z = 0 \quad (1), \quad -2y - 5z = 0 \quad (2).$$

From (2), $-2y = 5z \Rightarrow y = -\frac{5}{2}z$; take $z = k \Rightarrow \boxed{y = -\frac{5}{2}k}$. $\therefore$ From (1), $x = -3z - 2y = -3(k) - 2(-\frac{5}{2}k) = -3k + 5k \Rightarrow \boxed{x = 2k}$.

$\therefore$ The infinite solution is $x = 2k, y = -\frac{5}{2}k$ and $z = k$, and k takes any value. ■

7 Applications: modal analysis, mesh analysis and other models

7.1 Modal analysis of electrical circuits

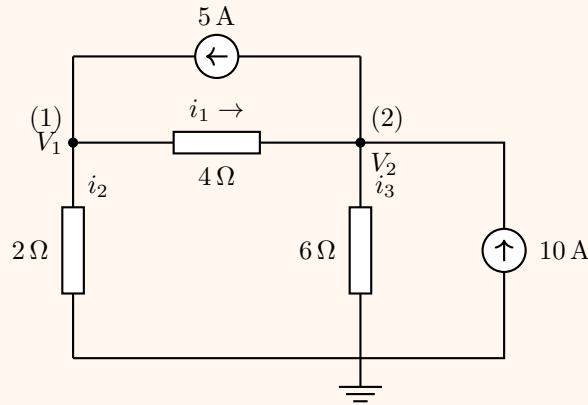
Modal analysis is the study of dynamic properties of a system. In modal analysis, consider an electric circuit which does not contain voltage sources, and our aim is to find the node voltages using the given conditions. We require two important laws of electric circuits.

Result 7.1 (Laws of electric circuits)

- (1) **Kirchhoff's law (current law at a node):** the algebraic sum of the currents entering a node is zero (e.g. if i_1 enters and i_2, i_3 leave, $i_1 - i_2 - i_3 = 0$, i.e. $i_1 = i_2 + i_3$).
- (2) **Ohm's law:** Current $i = \frac{V}{R} = \frac{\text{Potential difference of voltages}}{\text{Resistance}}$.

Problem 7.1

Calculate the voltages in the two nodes in the following circuit, by solving the resultant simultaneous equations by the matrix method (Gauss elimination, i.e. row-echelon form).



(Node 1: current 5 A enters from the source branch and splits into i_1 through 4Ω and i_2 through 2Ω . Node 2: i_1 and 10 A enter, i_3 through 6Ω and 5 A leave.)

Solution. Based on the circuit, we can represent the currents in the figure. Let V_1 and V_2 be the voltages at the two nodes 1 and 2 respectively.

At node 1 (by Kirchhoff's law): $i_1 + i_2 = 5$. By Ohm's law,

$$\frac{V_1 - V_2}{4} + \frac{V_1 - 0}{2} = 5 \Rightarrow V_1 - V_2 + 2V_1 = 20 \Rightarrow \boxed{3V_1 - V_2 = 20} \quad (1)$$

At node 2: $i_1 + 10 = i_3 + 5$ (Kirchhoff's law) $\Rightarrow i_1 - i_3 = -5$

$$\Rightarrow \frac{V_1 - V_2}{4} - \left(\frac{V_2 - 0}{6} \right) = -5 \quad (\text{Ohm's law}) \Rightarrow 3V_1 - 3V_2 - 2V_2 = -60 \Rightarrow \boxed{3V_1 - 5V_2 = -60} \quad (2)$$

$\therefore$ We have the following algebraic equations at the two nodes: $3V_1 - V_2 = 20$, $3V_1 - 5V_2 = -60$. These equations can be written as

$$\begin{pmatrix} 3 & -1 \\ 3 & -5 \end{pmatrix} \begin{pmatrix} V_1 \\ V_2 \end{pmatrix} = \begin{pmatrix} 20 \\ -60 \end{pmatrix}, \quad \text{i.e. of the form } AV = B, \quad A = \begin{pmatrix} 3 & -1 \\ 3 & -5 \end{pmatrix}, \quad V = \begin{pmatrix} V_1 \\ V_2 \end{pmatrix}, \quad B = \begin{pmatrix} 20 \\ -60 \end{pmatrix}.$$

Then the augmented matrix is $(A, B) = \left(\begin{array}{cc|c} 3 & -1 & 20 \\ 3 & -5 & -60 \end{array} \right)$. Apply $R_2 \rightarrow R_2 - R_1$:

$$(A, B) \sim \left(\begin{array}{cc|c} 3 & -1 & 20 \\ 0 & -4 & -80 \end{array} \right) \text{ is the row-echelon form,}$$

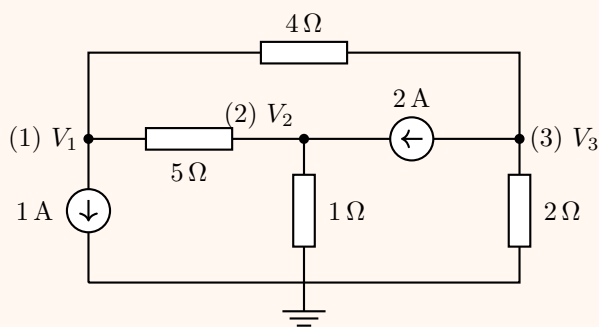
and it can again be written as $AV = B$:

$$\begin{pmatrix} 3 & -1 \\ 0 & -4 \end{pmatrix} \begin{pmatrix} V_1 \\ V_2 \end{pmatrix} = \begin{pmatrix} 20 \\ -80 \end{pmatrix} \Rightarrow 3V_1 - V_2 = 20, \quad -4V_2 = -80 \Rightarrow \boxed{V_2 = 20} \text{ and } 3V_1 = 40.$$

$\Rightarrow$ The solution is $V_1 = \frac{40}{3}$ volts, $V_2 = 20$ volts. ■

Problem 7.2

Determine the voltages in the three nodes of the following circuit.



Solution. At node 1: $i_1 + i_2 + 1 = 0$ (i_1 through 5Ω to node 2, i_2 through 4Ω to node 3):

$$\frac{V_1 - V_2}{5} + \frac{V_1 - V_3}{4} + 1 = 0 \Rightarrow 4V_1 - 4V_2 + 5V_1 - 5V_3 = -20 \Rightarrow \boxed{9V_1 - 4V_2 - 5V_3 = -20} \quad (1)$$

At node 2: $i_1 + 2 = i_4$:

$$\frac{V_1 - V_2}{5} + 2 = \frac{V_2 - 0}{1} \Rightarrow V_1 - V_2 - 5V_2 = -10 \Rightarrow \boxed{V_1 - 6V_2 = -10} \quad (2)$$

At node 3: $i_2 = 2 + i_3$:

$$\frac{V_1 - V_3}{4} = 2 + \frac{V_3 - 0}{2} \Rightarrow V_1 - V_3 - 2V_3 = 8 \Rightarrow \boxed{V_1 - 3V_3 = 8} \quad (3)$$

Equations (1), (2) and (3) can be written as $AV = B$, i.e.

$$\begin{pmatrix} 9 & -4 & -5 \\ 1 & -6 & 0 \\ 1 & 0 & -3 \end{pmatrix} \begin{pmatrix} V_1 \\ V_2 \\ V_3 \end{pmatrix} = \begin{pmatrix} -20 \\ -10 \\ 8 \end{pmatrix}. \quad \text{Then } (A, B) = \left(\begin{array}{ccc|c} 9 & -4 & -5 & -20 \\ 1 & -6 & 0 & -10 \\ 1 & 0 & -3 & 8 \end{array} \right).$$

Apply $R_1 \leftrightarrow R_3$:

$$(A, B) \sim \left(\begin{array}{ccc|c} 1 & 0 & -3 & 8 \\ 1 & -6 & 0 & -10 \\ 9 & -4 & -5 & -20 \end{array} \right). \quad \text{Apply } R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - 9R_1: (A, B) \sim \left(\begin{array}{ccc|c} 1 & 0 & -3 & 8 \\ 0 & -6 & 3 & -18 \\ 0 & -4 & 22 & -92 \end{array} \right)$$

$(-5 + 27 = 22, -20 - 72 = -92)$. Apply $R_2 \rightarrow -\frac{1}{6}R_2$:

$$(A, B) \sim \left(\begin{array}{ccc|c} 1 & 0 & -3 & 8 \\ 0 & 1 & -\frac{1}{2} & 3 \\ 0 & -4 & 22 & -92 \end{array} \right). \quad \text{Apply } R_3 \rightarrow R_3 + 4R_2: (A, B) \sim \left(\begin{array}{ccc|c} 1 & 0 & -3 & 8 \\ 0 & 1 & -\frac{1}{2} & 3 \\ 0 & 0 & 20 & -80 \end{array} \right)$$

$(22 - 2 = 20, -92 + 12 = -80)$, which is the row-echelon form. Expanding in the form $AV = B$:

$$V_1 - 3V_3 = 8 \quad (1), \quad V_2 - \frac{1}{2}V_3 = 3 \quad (2), \quad 20V_3 = -80 \quad (3).$$

From (3), $\boxed{V_3 = -4}$ volts; (2) $\Rightarrow V_2 + 2 = 3 \Rightarrow \boxed{V_2 = 1}$ volt; (1) $\Rightarrow V_1 = 8 + 3V_3 = 8 + 3(-4) \Rightarrow \boxed{V_1 = -4}$ volts.

The solution is $V_1 = -4, V_2 = 1, V_3 = -4$ volts. ■

7.2 Balancing a chemical equation

Problem 7.3

Balance the chemical equation $x_1 \text{C}_2\text{H}_6 + x_2 \text{O}_2 \rightarrow x_3 \text{CO}_2 + x_4 \text{H}_2\text{O}$.

Solution. Equating the number of atoms of each element on both sides: carbon $2x_1 = x_3$; hydrogen $6x_1 = 2x_4$; oxygen $2x_2 = 2x_3 + x_4$. We have

$$(A, B) = \left(\begin{array}{cccc|c} 2 & 0 & -1 & 0 & 0 \\ 6 & 0 & 0 & -2 & 0 \\ 0 & 2 & -2 & -1 & 0 \end{array} \right) \quad (\text{columns } x_1, x_2, x_3, x_4).$$

Apply $R_2 \rightarrow R_2 - 3R_1$:

$$(A, B) \sim \left(\begin{array}{cccc|c} 2 & 0 & -1 & 0 & 0 \\ 0 & 0 & 3 & -2 & 0 \\ 0 & 2 & -2 & -1 & 0 \end{array} \right). \text{ Apply } R_2 \leftrightarrow R_3 : (A, B) \sim \left(\begin{array}{cccc|c} 2 & 0 & -1 & 0 & 0 \\ 0 & 2 & -2 & -1 & 0 \\ 0 & 0 & 3 & -2 & 0 \end{array} \right),$$

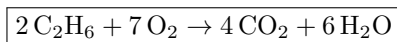
which is the row-echelon form. Expanding the above (A, B) into $AX = B$ form, we get

$$2x_1 - x_3 = 0 \quad (1), \quad 2x_2 - 2x_3 - x_4 = 0 \quad (2), \quad 3x_3 - 2x_4 = 0 \quad (3).$$

From (3), $3x_3 = 2x_4$; take $x_4 = k \Rightarrow x_3 = \frac{2}{3}k$. (1) $\Rightarrow x_1 = \frac{1}{2}x_3 \Rightarrow x_1 = \frac{1}{3}k$. (2) $\Rightarrow x_2 = \frac{1}{2}(2x_3 + x_4) = \frac{1}{2}(\frac{4}{3}k + k) \Rightarrow x_2 = \frac{7}{6}k$.

I.e. we have $x_1 = \frac{k}{3}$, $x_2 = \frac{7k}{6}$, $x_3 = \frac{2k}{3}$ and $x_4 = k$. But we have to fix a positive integer value for k to balance the chemical equation. $\therefore$ The least positive integer value is $k = 6$, and we get $x_1 = 2$; $x_2 = 7$; $x_3 = 4$ and $x_4 = 6$.

$\therefore$ The required balanced chemical equation is



■

7.3 Traffic flow problem

Problem 7.4

In a network of one-way streets with junctions A, B, C, D , the traffic (vehicles per hour, vph) entering and leaving each junction gives: at A , traffic in $= x + y$, traffic out $= 190 + 241 = 431$; at B , traffic in $= 105 + 150 = 255$, traffic out $= x + w$; similarly at C : $w + z = 340$ and at D : $y + z = 516$. Determine the traffic flows x, y, z, w .

Solution. At junction A: $x + y = 431$; at junction B: $x + w = 255$; at junction C: $w + z = 340$; at junction D: $y + z = 516$. I.e. we have

$$x + y + 0z + 0w = 431, \quad x + 0y + 0z + w = 255, \quad 0x + 0y + z + w = 340, \quad 0x + y + z + 0w = 516$$

$$\Rightarrow (A, B) = \left(\begin{array}{cccc|c} 1 & 1 & 0 & 0 & 431 \\ 1 & 0 & 0 & 1 & 255 \\ 0 & 0 & 1 & 1 & 340 \\ 0 & 1 & 1 & 0 & 516 \end{array} \right).$$

Apply $R_2 \rightarrow R_2 - R_1$ ($255 - 431 = -176$):

$$(A, B) \sim \left(\begin{array}{cccc|c} 1 & 1 & 0 & 0 & 431 \\ 0 & -1 & 0 & 1 & -176 \\ 0 & 0 & 1 & 1 & 340 \\ 0 & 1 & 1 & 0 & 516 \end{array} \right). \text{ Apply } R_4 \rightarrow R_4 + R_2 : (A, B) \sim \left(\begin{array}{cccc|c} 1 & 1 & 0 & 0 & 431 \\ 0 & -1 & 0 & 1 & -176 \\ 0 & 0 & 1 & 1 & 340 \\ 0 & 0 & 1 & 1 & 340 \end{array} \right).$$

Apply $R_4 \rightarrow R_4 - R_3$:

$$(A, B) \sim \left(\begin{array}{cccc|c} 1 & 1 & 0 & 0 & 431 \\ 0 & -1 & 0 & 1 & -176 \\ 0 & 0 & 1 & 1 & 340 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right),$$

which is the row-echelon form. If we expand the above (A, B) in the form $AX = B$, we get

$$x + y = 431 \quad (1), \quad -y + w = -176 \quad (2), \quad z + w = 340 \quad (3).$$

From (3), $z = 340 - w$, i.e. in this problem w is a free variable (the traffic on the interchange road). If we assume the traffic density $w = 100$ vph, then $z = 240$ vph, and from (2) and (1)

$$y = 276 \text{ vph}, \quad x = 155 \text{ vph}.$$

■

7.4 Electric circuit problems involving voltage sources (mesh analysis)

Problem 7.5

Solve for the mesh currents i_1, i_2, i_3 whose augmented matrix (from Kirchhoff's laws) is

$$(A, B) = \left(\begin{array}{ccc|c} 1 & -1 & -1 & 0 \\ 10 & 20 & 0 & -12 \\ 0 & -20 & 10 & 0 \end{array} \right).$$

Solution. Apply $R_2 \rightarrow R_2 - 10R_1$:

$$(A, B) \sim \left(\begin{array}{ccc|c} 1 & -1 & -1 & 0 \\ 0 & 30 & 10 & -12 \\ 0 & -20 & 10 & 0 \end{array} \right). \text{ Apply } R_3 \rightarrow R_3 + \frac{2}{3}R_2 : (A, B) \sim \left(\begin{array}{ccc|c} 1 & -1 & -1 & 0 \\ 0 & 30 & 10 & -12 \\ 0 & 0 & \frac{50}{3} & -8 \end{array} \right),$$

which is the row-echelon form of the matrix. Expanding this (A, B) into $Ai = B$ form: $\frac{50}{3}i_3 = -8 \Rightarrow i_3 = -0.48$; $30i_2 + 10i_3 = -12 \Rightarrow 30i_2 = -12 + 4.8 \Rightarrow i_2 = -0.24$; $i_1 = i_2 + i_3 = -0.72$. (The negative signs indicate that the actual directions of the currents are opposite to those assumed.) ■

7.5 Population growth model

Problem 7.6 (Example)

A rabbit population has three age classes. The age transition matrix is $A = \begin{pmatrix} 0 & 6 & 8 \\ 0.5 & 0 & 0 \\ 0 & 0.5 & 0 \end{pmatrix}$ and the current age distribution vector is $X_0 = \begin{pmatrix} 24 \\ 24 \\ 20 \end{pmatrix}$ (first, second and third age classes). Find the age distribution after one year, and the stable age distribution vector.

Solution. The number of rabbits in each age class in the next year is given by

$$X_1 = AX_0 = \begin{pmatrix} 0 & 6 & 8 \\ 0.5 & 0 & 0 \\ 0 & 0.5 & 0 \end{pmatrix} \begin{pmatrix} 24 \\ 24 \\ 20 \end{pmatrix} = \begin{pmatrix} 0 + 6(24) + 8(20) \\ 0.5(24) + 0 + 0 \\ 0 + 0.5(24) + 0 \end{pmatrix} = \begin{pmatrix} 304 \\ 12 \\ 12 \end{pmatrix} \begin{array}{l} \leftarrow \text{1st age class} \\ \leftarrow \text{2nd age class} \\ \leftarrow \text{3rd age class} \end{array}$$

($144 + 160 = 304$), i.e. X_1 gives the age distribution vector for the number of rabbits in the first year in each age class. Similarly $X_2 = AX_1$ gives the number of rabbits in the 2nd year in each age class, $X_3 = AX_2$ gives the number of rabbits in the 3rd year in each age class, and so on.

Remark. Suppose we wish to identify the stable age distribution vector in each year; we must have age distribution vectors like $X_1 = AX_0 = \lambda_1 X_0$, $X_2 = AX_1 = \lambda_2 X_0$, ..., $X_n = AX_{n-1} = \lambda_n X_0$. This is like the eigen value–eigen vector equation $AX = \lambda X$. $\therefore$ To identify the stable age distribution vector, first find the eigen values of the matrix A and then determine the eigen vector for the respective eigen value. This eigen vector will provide the stable age distribution vector in each year and in each age class.

Stable age distribution. Given $A = \begin{pmatrix} 0 & 6 & 8 \\ 0.5 & 0 & 0 \\ 0 & 0.5 & 0 \end{pmatrix}$. The characteristic equation is

$$\lambda^3 - (0)\lambda^2 + (0 + 0 + (-3))\lambda - |A| = 0, \quad |A| = 0 - 6(0) + 8(0.25) = 2 \Rightarrow \lambda^3 - 3\lambda - 2 = 0.$$

When $\lambda = -1$: $-1 + 3 - 2 = 0 \Rightarrow \boxed{\lambda = -1}$ is one root. By synthetic division

$$\begin{array}{r|rrrr} -1 & 1 & 0 & -3 & -2 \\ & & -1 & 1 & 2 \\ \hline & 1 & -1 & -2 & 0 \end{array} \Rightarrow \lambda^2 - \lambda - 2 = 0 \Rightarrow (\lambda - 2)(\lambda + 1) = 0.$$

$\Rightarrow$ Eigen values are $\lambda = -1, -1, 2$. To determine the stable growth vector, we can use the positive λ value, i.e. $\lambda = 2$. Then we have the eigen vector from $AX = \lambda X$:

$$\begin{pmatrix} 0 & 6 & 8 \\ 0.5 & 0 & 0 \\ 0 & 0.5 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 2 \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \Rightarrow -2x_1 + 6x_2 + 8x_3 = 0 \text{ (1)}, \quad 0.5x_1 - 2x_2 = 0 \text{ (2)}, \quad 0.5x_2 - 2x_3 = 0 \text{ (3)}.$$

By CMR on (1) and (2):

$$\frac{x_1}{0+16} = \frac{-x_2}{0-4} = \frac{x_3}{4-3} = k \Rightarrow \frac{x_1}{16} = \frac{x_2}{4} = \frac{x_3}{1} = k.$$

When $k = 1$, the stable growth age distribution vector is $X_1 = \begin{pmatrix} 16 \\ 4 \\ 1 \end{pmatrix}$. Check: $X_2 = AX_1 = \begin{pmatrix} 0 & 6 & 8 \\ 0.5 & 0 & 0 \\ 0 & 0.5 & 0 \end{pmatrix} \begin{pmatrix} 16 \\ 4 \\ 1 \end{pmatrix} = \begin{pmatrix} 32 \\ 8 \\ 2 \end{pmatrix} = 2X_1$; if we find $X_2 = AX_1$ we get again a multiple of the vector X_1 .

$\therefore$ The stable age distribution vector in each age class is $X = \begin{pmatrix} 16 \\ 4 \\ 1 \end{pmatrix}$ (1st, 2nd, 3rd age classes), i.e. of the form $16 : 4 : 1$. ■

Assignment – II

- (1) Prove that the set of vectors $X_1 = (1, -2, 4, 1)$, $X_2 = (1, 0, 6, -5)$, $X_3 = (2, -3, 9, -1)$ and $X_4 = (2, -5, 7, 5)$ are linearly dependent and find their relationship.
- (2) Show that the vectors $X_1 = (1, 2, -1, 0)$, $X_2 = (1, 3, 1, 2)$, $X_3 = (4, 2, 1, 0)$ and $X_4 = (6, 1, 0, 1)$ are linearly independent.
- (3) Find the value of λ if the equations $x + y - z = 10$, $x - y + 2z = 20$ and $\lambda x - y + 4z = 30$ have a unique solution.
- (4) Test the consistency of the equations $3x + 5y - 2z = 1$, $x - y + 4z = 7$, $-6x - 2y + 5z = 9$, $7x - 3y + z = 4$ and solve if they are consistent.
- (5) Prove that the eigen values of a real symmetric matrix are real. Also prove that the eigen vectors corresponding to distinct eigen values of a real symmetric matrix are orthogonal.
- (6) Prove that the eigen values of $-3A^{-1}$ are the same as those of $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$.
- (7) Find the sum of the eigen values of the inverse of $A = \begin{pmatrix} 3 & 0 & 0 \\ 8 & 4 & 0 \\ 6 & 2 & 5 \end{pmatrix}$.
- (8) Two eigen values of the matrix $A = \begin{pmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{pmatrix}$ are equal to 1 each. Find the third eigen value.
- (9) The product of two eigen values of the matrix $A = \begin{pmatrix} 6 & -2 & 2 \\ -3 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$ is 16. Find the third eigen value.
- (10) Show that the eigen vectors of $A = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$ are $\begin{pmatrix} 1 \\ i \end{pmatrix}$ and $\begin{pmatrix} 1 \\ -i \end{pmatrix}$.
- (11) Verify that the matrix $A = \frac{1}{3} \begin{pmatrix} -1 & 2 & -2 \\ -2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix}$ is orthogonal and that its eigen values are of unit modulus.

- (12) What can you infer about the matrix $A = \begin{pmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{pmatrix}$ from its eigen values?
- (13) Reduce the quadratic form $Q : 2x^2 + y^2 + z^2 + 2xy - 2xz - 4yz = 6$ to canonical form by an orthogonal transformation [or: rotate the quadratic surface Q through a rotation angle θ , find its equation w.r.t. the new principal axes x', y', z' and also write the rotation angle θ]. Also find the rank, index, signature and nature of the quadratic form.
- (14) Diagonalize the matrix $A = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ -4 & 4 & 3 \end{pmatrix}$ by similarity transformation. Find also the 4th power of this matrix using the diagonal matrix ($A^4 = BD^4B^{-1}$).
- (15) Diagonalize the matrix $A = \begin{pmatrix} 2 & -2 & 2 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{pmatrix}$ if possible.